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Systems and methods for additive manufacturing of a golf club

Abstract

A golf club head that includes a body that has a topline, a sole, and a front face. The front face defines a rear surface that extends along a plane and the body defines a center of gravity plane. The golf club head also includes a lattice structure formed on a portion of the body, layer by layer, via an additive manufacturing process. The portion of the body is bounded by the plane, the center of gravity plane, and an intersection between the plane and the center of gravity plane.

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9539477	12/2016	Ripp et al.	N/A	N/A
9545544	12/2016	Jones et al.	N/A	N/A
9550096	12/2016	Parsons et al.	N/A	N/A
9550097	12/2016	Myers et al.	N/A	N/A

9555295	12/2016	Schweigert et al.	N/A	N/A
9561413	12/2016	Nielson et al.	N/A	N/A
9566479	12/2016	Albertsen et al.	N/A	N/A
9586105	12/2016	Westrum et al.	N/A	N/A
9597558	12/2016	Seluga et al.	N/A	N/A
9597561	12/2016	Seluga et al.	N/A	N/A
9604108	12/2016	Dunnell et al.	N/A	N/A
9610481	12/2016	Parsons et al.	N/A	N/A
9610482	12/2016	Burnett et al.	N/A	N/A
9610483	12/2016	Burnett et al.	N/A	N/A
9616303	12/2016	Wu et al.	N/A	N/A
9616304	12/2016	Deshmukh et al.	N/A	N/A
9623294	12/2016	Kingston et al.	N/A	N/A
9623301	12/2016	Bamber	N/A	N/A
9623302	12/2016	Myers et al.	N/A	N/A
9630069	12/2016	Foster et al.	N/A	N/A
9630070	12/2016	Parsons et al.	N/A	N/A
9630080	12/2016	Lanyi	N/A	N/A
9636553	12/2016	Myers	N/A	N/A
9636554	12/2016	Parsons et al.	N/A	N/A
9649540	12/2016	Parsons et al.	N/A	N/A
9649542	12/2016	Nicolette	N/A	N/A
9656131	12/2016	Burnett et al.	N/A	N/A
9662545	12/2016	Beach et al.	N/A	N/A
9662547	12/2016	Parsons et al.	N/A	N/A
9662551	12/2016	Akiyama et al.	N/A	N/A
9669270	12/2016	Schweigert et al.	N/A	N/A
9675849	12/2016	Beach et al.	N/A	N/A
9675853	12/2016	Parsons et al.	N/A	N/A
9675854	12/2016	Wang et al.	N/A	N/A
9675856	12/2016	Gibbs et al.	N/A	N/A
9675864	12/2016	Carter	N/A	N/A
9682295	12/2016	Dawson et al.	N/A	N/A
9682296	12/2016	Myers et al.	N/A	N/A
9682298	12/2016	Kingston et al.	N/A	N/A
9682299	12/2016	Tang et al.	N/A	N/A
9687701	12/2016	Seluga et al.	N/A	N/A
9687702	12/2016	Seluga et al.	N/A	N/A
9687705	12/2016	Boyd et al.	N/A	N/A
9694252	12/2016	Beach et al.	N/A	N/A
9694254	12/2016	Ashino	N/A	A63B 60/00
9694256	12/2016	Myers et al.	N/A	N/A
9694257	12/2016	Seluga et al.	N/A	N/A
9694259	12/2016	Trahan et al.	N/A	N/A
9694260	12/2016	Abbott et al.	N/A	N/A
9694261	12/2016	Nunez et al.	N/A	N/A
9694262	12/2016	Myers	N/A	N/A
9700763	12/2016	Harbert et al.	N/A	N/A
9700769	12/2016	Beach et al.	N/A	N/A
9707457	12/2016	Mata et al.	N/A	N/A

9707459	12/2016	Myers	N/A	N/A
9713752	12/2016	Myers et al.	N/A	N/A
9717961	12/2016	Cameron	N/A	N/A
9724577	12/2016	Kingston et al.	N/A	N/A
9731173	12/2016	Bezilla et al.	N/A	N/A
9731175	12/2016	Myers et al.	N/A	N/A
9731178	12/2016	Myers	N/A	N/A
9757629	12/2016	Seluga et al.	N/A	N/A
9757630	12/2016	Mata et al.	N/A	N/A
9764204	12/2016	Beach et al.	N/A	N/A
9764208	12/2016	Parsons et al.	N/A	N/A
9764209	12/2016	Evans	N/A	N/A
9770632	12/2016	Boyd et al.	N/A	N/A
9776051	12/2016	Abbott	N/A	N/A
9776058	12/2016	Seluga et al.	N/A	N/A
9776060	12/2016	Hall et al.	N/A	N/A
9782642	12/2016	DeMille et al.	N/A	N/A
9782643	12/2016	Parsons et al.	N/A	N/A
9782645	12/2016	Dolezel	N/A	N/A
9782646	12/2016	Seluga et al.	N/A	N/A
9789372	12/2016	Beach et al.	N/A	N/A
9795842	12/2016	Parsons et al.	N/A	N/A
9795843	12/2016	Parsons et al.	N/A	N/A
9795844	12/2016	Dacey et al.	N/A	N/A
9795846	12/2016	Sargent et al.	N/A	N/A
9796131	12/2016	Parsons et al.	N/A	N/A
9802087	12/2016	Schweigert et al.	N/A	N/A
9802372	12/2016	Stites et al.	N/A	N/A
9808680	12/2016	Myers et al.	N/A	N/A
9814945	12/2016	Parsons et al.	N/A	N/A
9814947	12/2016	Seluga et al.	N/A	N/A
9814952	12/2016	Parsons et al.	N/A	N/A
9814953	12/2016	Sargent et al.	N/A	N/A
9814954	12/2016	Westrum et al.	N/A	N/A
9821199	12/2016	Seluga et al.	N/A	N/A
9821200	12/2016	Parsons et al.	N/A	N/A
9821201	12/2016	Parsons et al.	N/A	N/A
9821521	12/2016	Stites et al.	N/A	N/A
9827469	12/2016	Seluga et al.	N/A	N/A
9827479	12/2016	Demkowski et al.	N/A	N/A
9833667	12/2016	Parsons et al.	N/A	N/A
9833669	12/2016	Myers et al.	N/A	N/A
9839818	12/2016	Jertson et al.	N/A	N/A
9839822	12/2016	Becktor et al.	N/A	N/A
9839823	12/2016	Ehlers	N/A	N/A
9844707	12/2016	Davis	N/A	N/A
9844710	12/2016	Parsons et al.	N/A	N/A
9849353	12/2016	Beach et al.	N/A	N/A
9849356	12/2016	Roach et al.	N/A	N/A
9849359	12/2016	Gibbs et al.	N/A	N/A

9855473	12/2017	Beach et al.	N/A	N/A
9855476	12/2017	Seluga et al.	N/A	N/A
9861867	12/2017	Parsons et al.	N/A	N/A
9868036	12/2017	Kleinert et al.	N/A	N/A
9873027	12/2017	Hamada	N/A	N/A
9878218	12/2017	Parsons et al.	N/A	N/A
9878220	12/2017	Parsons et al.	N/A	N/A
9878223	12/2017	Foster et al.	N/A	N/A
9884230	12/2017	Snyder	N/A	N/A
9889349	12/2017	Seluga et al.	N/A	N/A
9895582	12/2017	Schweigert et al.	N/A	N/A
9895583	12/2017	Parsons et al.	N/A	N/A
9908011	12/2017	Larson et al.	N/A	N/A
9908012	12/2017	Larson et al.	N/A	N/A
9908014	12/2017	Wester et al.	N/A	N/A
9908016	12/2017	Seluga et al.	N/A	N/A
9908017	12/2017	Seluga et al.	N/A	N/A
9914025	12/2017	Larson et al.	N/A	N/A
9914027	12/2017	Harbert et al.	N/A	N/A
9914029	12/2017	Parsons et al.	N/A	N/A
9914040	12/2017	Carter	N/A	N/A
9919190	12/2017	Beach et al.	N/A	N/A
9925427	12/2017	Bischmann et al.	N/A	N/A
9931549	12/2017	Seluga et al.	N/A	N/A
9931550	12/2017	Seluga et al.	N/A	N/A
9937387	12/2017	Wang et al.	N/A	N/A
9937392	12/2017	Myers et al.	N/A	N/A
9950222	12/2017	Albertsen et al.	N/A	N/A
9950223	12/2017	Burnett et al.	N/A	N/A
9956460	12/2017	Burnett et al.	N/A	N/A
9956462	12/2017	Nunez et al.	N/A	N/A
9962583	12/2017	Abbott et al.	N/A	N/A
9962584	12/2017	Nielson et al.	N/A	N/A
9968834	12/2017	Seluga et al.	N/A	N/A
9975011	12/2017	Wester et al.	N/A	N/A
9975015	12/2017	Ripp et al.	N/A	N/A
9981160	12/2017	Parsons et al.	N/A	N/A
9981161	12/2017	Abbott	N/A	N/A
9981165	12/2017	Goudarzi et al.	N/A	N/A
9981167	12/2017	Seluga et al.	N/A	N/A
9981170	12/2017	Myers	N/A	N/A
9987523	12/2017	Beach et al.	N/A	N/A
9987526	12/2017	Parsons et al.	N/A	N/A
9987527	12/2017	Myers et al.	N/A	N/A
9987531	12/2017	Cameron	N/A	N/A
9999814	12/2017	Parsons et al.	N/A	N/A
10004955	12/2017	Henrikson et al.	N/A	N/A
10004958	12/2017	Tang et al.	N/A	N/A
10010770	12/2017	Parsons et al.	N/A	N/A
10010771	12/2017	Seluga et al.	N/A	N/A

10016662	12/2017	Wester et al.	N/A	N/A
10022601	12/2017	Petersen et al.	N/A	N/A
10022602	12/2017	Kingston et al.	N/A	N/A
10029158	12/2017	Parsons et al.	N/A	N/A
10029159	12/2017	Parsons et al.	N/A	N/A
10029160	12/2017	Stites et al.	N/A	N/A
10035049	12/2017	Nielson et al.	N/A	N/A
10046212	12/2017	Sargent et al.	N/A	N/A
10052529	12/2017	Abbott et al.	N/A	N/A
10052532	12/2017	Parsons et al.	N/A	N/A
10058748	12/2017	Dolezel et al.	N/A	N/A
10058749	12/2017	Beach et al.	N/A	N/A
10065085	12/2017	Seluga et al.	N/A	N/A
10065086	12/2017	Seluga et al.	N/A	N/A
10065091	12/2017	Hall et al.	N/A	N/A
10071290	12/2017	Boyd et al.	N/A	N/A
10071292	12/2017	Wu et al.	N/A	N/A
10071293	12/2017	Green	N/A	N/A
10076688	12/2017	Harbert et al.	N/A	N/A
10076690	12/2017	Myers	N/A	N/A
10080932	12/2017	DeMille et al.	N/A	N/A
10080934	12/2017	Beach et al.	N/A	N/A
10086238	12/2017	Roach et al.	N/A	N/A
10086240	12/2017	Hoffman et al.	N/A	N/A
10086243	12/2017	Sheldon et al.	N/A	N/A
10086244	12/2017	Morin et al.	N/A	N/A
10092801	12/2017	Sheldon et al.	N/A	N/A
10099093	12/2017	Parsons et al.	N/A	N/A
10099095	12/2017	Myers	N/A	N/A
10112083	12/2017	Jones et al.	N/A	N/A
10112084	12/2017	Stokke et al.	N/A	N/A
10124219	12/2017	Sargent et al.	N/A	N/A
10124220	12/2017	Cameron	N/A	N/A
10124221	12/2017	Parsons et al.	N/A	N/A
10124224	12/2017	Solheim	N/A	A63B 53/0466
10137340	12/2017	Fay	N/A	N/A
10143899	12/2017	Schweigert et al.	N/A	N/A
10150017	12/2017	Boyd et al.	N/A	N/A
10155140	12/2017	Sargent et al.	N/A	N/A
10155143	12/2017	Wahl et al.	N/A	N/A
10159876	12/2017	Parsons et al.	N/A	N/A
10173105	12/2018	Myers et al.	N/A	N/A
10173110	12/2018	Foster et al.	N/A	N/A
10173116	12/2018	Beach et al.	N/A	N/A
10183202	12/2018	Harbert et al.	N/A	N/A
10195501	12/2018	Parsons et al.	N/A	N/A
10195502	12/2018	Ripp et al.	N/A	N/A
10201733	12/2018	Seluga et al.	N/A	N/A
10207162	12/2018	Deshmukh	N/A	N/A
10213659	12/2018	Parsons et al.	N/A	N/A

10213661	12/2018	Seluga et al.	N/A	N/A
10213662	12/2018	Seluga et al.	N/A	N/A
10213663	12/2018	Goudarzi et al.	N/A	N/A
10220270	12/2018	Beach et al.	N/A	N/A
10220273	12/2018	Dolezel et al.	N/A	N/A
10220275	12/2018	Hebreo	N/A	N/A
10232234	12/2018	Parsons et al.	N/A	N/A
10232235	12/2018	Parsons et al.	N/A	N/A
10238932	12/2018	Becktor et al.	N/A	N/A
10238933	12/2018	Seluga et al.	N/A	N/A
10245476	12/2018	Abbott et al.	N/A	N/A
10245477	12/2018	DeMille et al.	N/A	N/A
10245485	12/2018	Burnett et al.	N/A	N/A
10252119	12/2018	Harbert et al.	N/A	N/A
10252123	12/2018	Parsons et al.	N/A	N/A
10252124	12/2018	Trahan et al.	N/A	N/A
10252138	12/2018	Demkowski et al.	N/A	N/A
10258843	12/2018	Morales et al.	N/A	N/A
10258844	12/2018	Schweigert	N/A	N/A
10258845	12/2018	Parsons et al.	N/A	N/A
10258846	12/2018	Nunez et al.	N/A	N/A
10265590	12/2018	Parsons et al.	N/A	N/A
10279224	12/2018	Webb et al.	N/A	N/A
10279229	12/2018	Day	N/A	A63B 53/007
10279233	12/2018	Parsons et al.	N/A	N/A
10286266	12/2018	Wester et al.	N/A	N/A
10286267	12/2018	Nicolette	N/A	N/A
10286268	12/2018	Parsons et al.	N/A	N/A
10293220	12/2018	Schweigert et al.	N/A	N/A
10293221	12/2018	Parsons et al.	N/A	N/A
10293222	12/2018	Snyder	N/A	N/A
10293223	12/2018	Kii	N/A	N/A
10293224	12/2018	Norimura et al.	N/A	N/A
10293225	12/2018	Sargent et al.	N/A	N/A
10293226	12/2018	Hebreo et al.	N/A	N/A
10293229	12/2018	Parsons et al.	N/A	N/A
10293235	12/2018	Beno et al.	N/A	N/A
10293236	12/2018	Uhm	N/A	N/A
10293237	12/2018	Hilton et al.	N/A	N/A
10300348	12/2018	Morris	N/A	N/A
10300350	12/2018	Albertsen et al.	N/A	N/A
10300355	12/2018	Petersen et al.	N/A	N/A
10300356	12/2018	Mata et al.	N/A	N/A
10315080	12/2018	Parsons et al.	N/A	N/A
10322321	12/2018	Oldknow et al.	N/A	N/A
10322324	12/2018	Dolezel	N/A	N/A
10328317	12/2018	Wang et al.	N/A	N/A
10335645	12/2018	Parsons et al.	N/A	N/A
10335647	12/2018	Seluga et al.	N/A	N/A
10335650	12/2018	Myers	N/A	N/A

10335653	12/2018	Daraskavich	N/A	B23K 11/115
10343031	12/2018	Day	N/A	A63B 53/0433
10363464	12/2018	Cameron	N/A	N/A
10363466	12/2018	Petersen et al.	N/A	N/A
10369429	12/2018	Burnett et al.	N/A	N/A
10369434	12/2018	Stites et al.	N/A	N/A
10376754	12/2018	Parsons et al.	N/A	N/A
10384102	12/2018	Parsons et al.	N/A	N/A
10391370	12/2018	Tassistro et al.	N/A	N/A
10398951	12/2018	Golden et al.	N/A	N/A
10406407	12/2018	DeMille et al.	N/A	N/A
10406408	12/2018	Seluga et al.	N/A	N/A
10413784	12/2018	Beach et al.	N/A	N/A
10413787	12/2018	Parsons et al.	N/A	N/A
10420989	12/2018	Parsons et al.	N/A	N/A
10420990	12/2018	Parsons et al.	N/A	N/A
10420994	12/2018	Beach et al.	N/A	N/A
10427015	12/2018	Balest	N/A	N/A
10434384	12/2018	Mata et al.	N/A	N/A
10441855	12/2018	Parsons et al.	N/A	N/A
10441858	12/2018	Schweigert	N/A	N/A
10442144	12/2018	Stites et al.	N/A	N/A
10449428	12/2018	Parsons et al.	N/A	N/A
10456634	12/2018	Abbott et al.	N/A	N/A
10456637	12/2018	Becktor et al.	N/A	N/A
10456638	12/2018	Dolezel et al.	N/A	N/A
10463932	12/2018	Sargent et al.	N/A	N/A
10471310	12/2018	Brady	N/A	N/A
10478679	12/2018	Harbert et al.	N/A	N/A
10478680	12/2018	Schweigert et al.	N/A	N/A
10478682	12/2018	Jeda et al.	N/A	N/A
10478683	12/2018	Stokke et al.	N/A	N/A
10478684	12/2018	Parsons et al.	N/A	N/A
10486036	12/2018	Rollinson	N/A	N/A
10486039	12/2018	Kingston et al.	N/A	N/A
10493331	12/2018	Kroloff et al.	N/A	N/A
10500453	12/2018	Carter	N/A	N/A
10512825	12/2018	Roach et al.	N/A	N/A
10512829	12/2018	Parsons et al.	N/A	N/A
10518139	12/2018	Jones et al.	N/A	N/A
10518141	12/2018	Goudarzi et al.	N/A	N/A
10532255	12/2019	Wester et al.	N/A	N/A
10532256	12/2019	Sargent et al.	N/A	N/A
10532257	12/2019	Schweigert et al.	N/A	N/A
10532258	12/2019	Seluga et al.	N/A	N/A
10532264	12/2019	Paster	N/A	N/A
10537771	12/2019	Ripp et al.	N/A	N/A
10537772	12/2019	Stephens	N/A	N/A
10537773	12/2019	Nielson et al.	N/A	N/A
10543406	12/2019	Carter	N/A	N/A

10543407	12/2019	Parsons	N/A	A63B 53/0466
10556158	12/2019	Harbert et al.	N/A	N/A
10556160	12/2019	Burnett et al.	N/A	N/A
10561911	12/2019	Parsons et al.	N/A	N/A
10569144	12/2019	Nielson et al.	N/A	N/A
10576338	12/2019	Beach et al.	N/A	N/A
10576339	12/2019	Schweigert et al.	N/A	N/A
10583335	12/2019	Wester et al.	N/A	N/A
10583336	12/2019	Parsons et al.	N/A	N/A
10583338	12/2019	Jertson et al.	N/A	N/A
10583344	12/2019	Hilton et al.	N/A	N/A
10589154	12/2019	Seluga et al.	N/A	N/A
10596424	12/2019	Parsons et al.	N/A	N/A
10596425	12/2019	Parsons et al.	N/A	N/A
10603555	12/2019	Beach et al.	N/A	N/A
10610746	12/2019	Larson et al.	N/A	N/A
10617917	12/2019	Parsons	N/A	A63B 53/0466
10617918	12/2019	Parsons et al.	N/A	N/A
10617920	12/2019	Nunez et al.	N/A	N/A
10617921	12/2019	Busch et al.	N/A	N/A
10625127	12/2019	Golden et al.	N/A	N/A
10632349	12/2019	Parsons et al.	N/A	N/A
10632351	12/2019	Solheim et al.	N/A	N/A
10632352	12/2019	Lambeth et al.	N/A	N/A
10639524	12/2019	Penney et al.	N/A	N/A
10639526	12/2019	Snyder	N/A	N/A
10646755	12/2019	Kleinert et al.	N/A	N/A
10646756	12/2019	Beach et al.	N/A	N/A
10646758	12/2019	Schweigert	N/A	N/A
10653928	12/2019	Parsons et al.	N/A	N/A
10653931	12/2019	Philip et al.	N/A	N/A
10661132	12/2019	Wallin	N/A	N/A
10668340	12/2019	Lambeth et al.	N/A	N/A
10668351	12/2019	Demkowski et al.	N/A	N/A
10682557	12/2019	Ambrose	N/A	N/A
10688349	12/2019	Stubben et al.	N/A	N/A
10688352	12/2019	Harbert et al.	N/A	N/A
10688355	12/2019	Parsons et al.	N/A	N/A
10688368	12/2019	Kovarik et al.	N/A	N/A
10695623	12/2019	Parsons et al.	N/A	N/A
10695624	12/2019	Parsons et al.	N/A	N/A
10835789	12/2019	DeMille	N/A	A63B 53/0487
11331544	12/2021	DeMille	N/A	G06F 30/10
11484757	12/2021	DeMille	N/A	A63B 53/045
11497973	12/2021	DeMille	N/A	A63B 60/08
11511166	12/2021	Roach	N/A	A63B 53/0454
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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional of U.S. application Ser. No. 16/852,324, filed on Apr. 17, 2020, which is incorporated by reference herein in its entirety.

REFERENCE REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

(1) Not applicable.

SEQUENCE LISTING

(2) Not applicable.

BACKGROUND

1. Field of the Disclosure

(3) The present disclosure relates to golf clubs, and more specifically to a golf club head that is manufactured via 3D printing or another type of additive manufacturing technique.

2. Description of the Background of the Disclosure

(4) Different types of golf clubs (e.g., irons, drivers, fairway woods, utility irons, hybrid irons/woods, putters, etc.) are used to effect different types of shots, based on a golfer's location and ball lie when playing a hole on a golf course. Typically, conventional golf club heads are formed by a forging process, a casting process, a metal injection molding process, or a machined process (e.g., milling), and then machined, ground, and/or polished to a factory finish standard (e.g., dimensions, loft, lie, weight, offset, surface finish, aesthetics, etc.).

(5) Forging and casting processes require the use of a mold to provide the general shape or body of a golf club head, which significantly reduces the ability of a golf club head to be customized or tailored to a specific set of performance characteristics. For example, the mold used in a forging or casting process may generally define where a center of gravity (CG) will be arranged for the golf club head. Post-forming machining may be implemented to remove material and slightly alter a location of the CG, but the design of the mold is the limiting factor for CG location flexibility.

(6) In addition, conventional golf club head manufacturing processes are limited from a geometric perspective by not being able to readily produce club heads with complex geometries. For example, golf club heads with undercut or hollow constructions may be required to be formed via a casting mold, which places volume-based constraints on CG location. Further, the geometric limitations of conventional golf club head manufacturing processes do not enable club heads to be manufactured with properties that leverage advantages from various club head types. For example, a large-volume, hollow construction club head generally provides a higher moment of inertia (MOI) when compared to a low-volume, forged club head (e.g., muscle back), but the low-volume forged club head may provide more consistent launch conditions and distance variability. Conventional club head manufacturing processes are limited to leveraging one set of advantages based on the type of club head being manufactured.

(7) Therefore, a need exists for golf club heads that are modifiable or customizable, and that can leverage performance advantages from a variety of club head types in a single club head without the restrictions present in conventional golf club head manufacturing processes.

SUMMARY

(8) The present disclosure is directed to golf club heads constructed using 3D printing or another type of additive manufacturing technique.

(9) In some embodiments, the present disclosure provides a golf club head that includes a body, a solid portion, and a lattice structure. The body includes a topline, a sole, and an internal cavity arranged between the topline and the sole. The solid portion is arranged within the internal cavity and is fabricated from a solid material. The lattice structure is arranged within the internal cavity and is formed layer by layer via an additive manufacturing process. The lattice structure defines a lattice volume and the solid portion defines a solid volume. An orientation of the lattice structure between the topline and the sole and a volume ratio between the lattice volume and the solid volume define a location of a center of gravity defined by the body.

(10) In some embodiments, the present disclosure provides a golf club head including a body and a lattice structure. The body includes a topline, a sole, and a front face. The lattice structure is formed on a portion of the body layer by layer along a build plane via an additive manufacturing process. When the build plane is oriented parallel to a normal defined by the front face, a lattice build angle defined between a lattice plane and the build plane is greater than or equal to about 30 degrees.

(11) In some embodiments, the present disclosure provides a golf club head including a body and a lattice structure. The body includes a topline, a sole, and a front face. The front face defines a rear surface that extends along a plane and the body defines a solid center of gravity plane. The lattice structure is formed on a portion of the body layer by layer via an additive manufacturing process. The portion of the body is bounded by the plane, the solid center of gravity plane, and an intersection between the plane and the solid center of gravity plane.

(12) In some embodiments, the present disclosure provides a golf club head that includes a body and a lattice structure. The body includes an insert wall, a crown, a sole, a heel, and a toe. A head

cavity is defined by the crown, the sole, the heel, and the toe. The lattice structure is disposed within the head cavity and extends from the crown and the sole. The lattice structure is unitary with the body.

(13) In some embodiments, the present disclosure provides a golf club head that includes a body and a lattice structure. The body includes an insert wall, a crown, a sole, a heel, and a toe. A head cavity is disposed within the body. The lattice structure is disposed within the head cavity and includes segments that extend from the crown and the sole. The lattice structure and the hosel are unitary with the body.

(14) In some embodiments, the present disclosure provides a 3-D printed golf club head post-printed component that includes a body, one or more material deposits, and a lattice structure. The body includes an insert wall, a crown, a sole, a heel, and a toe. A head cavity is disposed within the body. The one or more material deposits extend from one or more of the body and the hosel. The lattice structure is disposed within the head cavity and extends from internal surfaces of the body. The lattice structure and the hosel are unitary with the body.

(15) In some embodiments, the present disclosure provides a process of manufacturing a golf club head including the step of generating, via an additive manufacturing process, a golf club head. Generating the golf club head includes the steps of printing a first material, layer by layer, along a first plane, and creating a first blow through aperture that allows air to pass from a front portion of the golf club head to a head cavity disposed within a rear portion of the golf club head. The process further includes the steps of blowing excess material out from within the head cavity using the first blow through aperture and removing excess material formed at one or more material deposits disposed along the golf club head.

(16) In some embodiments, the present disclosure provides a process of manufacturing a golf club head including the step of generating, via an additive manufacturing process, a golf club head. Generating the golf club head includes the steps of printing a first material, layer by layer, to create a body defining a sole, a toe portion, a medial portion, a heel portion, and a head cavity, and creating a first blow through aperture that allows air to pass from a front portion of the golf club head to a rear portion of the club head component. The process further includes blowing excess material out from within the head cavity using the first blow through aperture.

(17) In some embodiments, the present disclosure provides a process of manufacturing a golf club head including the step of generating, via an additive manufacturing process, a golf club head. Generating the golf club head includes the steps of printing a first material, layer by layer, along a first plane, and creating a first blow through aperture that allows air to pass from a front portion of the golf club head to a head cavity disposed within a rear portion of the golf club head. The process further includes sintering the golf club head by setting the golf club head into a furnace such that the golf club head is resting on a second plane that defines an angle of between 10 degrees and about 50 degrees with respect to the first plane.

(18) In some embodiments, the present disclosure provides a process of manufacturing a golf club head that includes the steps of forming, via an additive manufacturing process, a body of the golf club head by printing, layer by layer, a boundary that encloses a volume and is formed by at least one layer, and sintering the body to form a solid material within the volume.

(19) In some embodiments, the present disclosure provides a process of manufacturing a golf club head that includes the steps of forming, via an additive manufacturing process, a body of the golf club head. Forming the body of the golf club head includes the steps of creating a cavity arranged within the body, printing a plug within the cavity that is separated from internal surfaces defined by the cavity, removing excess material within the cavity, moving the plug to a desired location within the cavity, filling the cavity with a filler material.

(20) In some embodiments, the present disclosure provides a process of manufacturing a face insert of a golf club head that includes the steps of forming via an additive manufacturing process, a mold insert, creating a mold from the mold insert formed via the additive manufacturing process, and

molding an insert from the mold insert. The mold insert includes a lattice structure or a ribbed structure protruding therefrom.

(21) In some embodiments, the present disclosure provides a process of manufacturing a golf club head that includes the steps of forming, via an additive manufacturing process, a body of the golf club head, arranging the body on a sintering support including a face surface and a hosel surface, and sintering the body of the golf club head.

(22) In some embodiments, the present disclosure provides a golf club head that includes a body formed layer by layer and having a topline, a sole, and an internal cavity arranged between the topline and the sole. The golf club head further includes at least one aperture formed through at least one of a hosel extending from the body, a rear surface of the body, and a toe portion of the body. The at least one aperture is configured to form a flow path that extends along the internal cavity and the at least one aperture.

(23) In some embodiments, the present disclosure provides a sintering support for a golf club head. The golf club head includes a front face and a hosel. The sintering support includes a face surface, a hosel surface that extends at an angle from one side of the face surface, and a support wall that extends from a side of the face surface opposite to the hosel surface.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a top, front, and left side isometric view of a putter-type club head in accordance with the present disclosure;

(2) FIG. 2 is a top, rear, and right side of the putter-type club head of FIG. 1;

(3) FIG. 3 is a front elevational view of the putter-type club head of FIG. 1;

(4) FIG. 4 is a rear elevational view of the putter-type club head of FIG. 1;

(5) FIG. 5 is left or toe side elevational view of the putter-type club head of FIG. 1;

(6) FIG. 6 is a right or heel side elevational view of the putter-type club head of FIG. 1;

(7) FIG. 7 is a top plan view of the putter-type club head of FIG. 1;

(8) FIG. 8 is a bottom plan view of the putter-type club head of FIG. 1;

(9) FIG. 9 is a top, front, and left side isometric view of the putter-type club head of FIG. 1 in a post 3D-printed state and a post-printed state;

(10) FIG. 10 is a bottom, rear, and right side isometric view of the post-printed putter-type club head of FIG. 9;

(11) FIG. 11 is a front elevational view of the post-printed putter-type club head of FIG. 9;

(12) FIG. 12 is a left side elevational view of the post-printed putter-type club head of FIG. 9;

(13) FIG. 13 is a rear elevational view of the post-printed putter-type club head of FIG. 9;

(14) FIG. 14 is a top, rear, and right side cross-sectional view taken through line 14-14 of FIG. 5;

(15) FIG. 15 is another top, rear, and right side cross-sectional view taken through line 15-15 of FIG. 5;

(16) FIG. 16 is still another top, rear, and right side cross-sectional view taken through line 16-16 of FIG. 5;

(17) FIG. 17 is a side cross sectional view taken through line 17-17 of FIG. 3;

(18) FIG. 18 is a side cross sectional view taken through line 18-18 of FIG. 3;

(19) FIG. 19 is a side cross-sectional view taken through line 19-19 of FIG. 11;

(20) FIG. 20 is a side cross-sectional view taken through line 20-20 of FIG. 3;

(21) FIG. 21 is a top, front, and left side view of a face insert that is insertable into the face cavity shown in the post-printed putter-type club of FIGS. 9-13;

(22) FIG. 22 is a side elevational view of the face insert of FIG. 21;

(23) FIG. 23 is a top, rear, and left side isometric view taken through line 23-23 of FIG. 4;

(24) FIG. 24 is a rear view of an iron-type golf club head according to one aspect of the present disclosure;

(25) FIG. 25 is a front view of the iron-type golf club head of FIG. 24;

(26) FIG. 26 is a rear view of the iron-type golf club head of FIG. 24 with an external shell transparent and a lattice structure extending over an internal cavity;

(27) FIG. 27 is a cross-sectional view of the iron-type golf club head of FIG. 26 taken along line 27-27;

(28) FIG. 28 is a rear view of the iron-type golf club head of FIG. 24 with an external shell transparent and a lattice structure extending over a portion of an internal cavity adjacent to a sole;

(29) FIG. 29 is a cross-sectional view of the iron-type golf club head of FIG. 28 taken along line 29-29;

(30) FIG. 30 is a rear view of the iron-type golf club head of FIG. 24 with an external shell transparent and a lattice structure extending over a portion of an internal cavity adjacent to a topline;

(31) FIG. 31 is a cross-sectional view of the iron-type golf club head of FIG. 30 taken along line 31-31;

(32) FIG. 32 is a top, front, right isometric view of an iron-type golf club head including a lattice structure with a gap formed between the lattice structure and a face insert with the face insert hidden;

(33) FIG. 33 is a cross-sectional view of the iron-type golf club head of FIG. 32 taken along line 33-33;

(34) FIG. 34 is a cross-sectional view of the iron-type golf club head of FIG. 32 taken along line 34-34 illustrating with a weight bar;

(35) FIG. 35 is a cross-sectional view of the iron-type golf club head of FIG. 34 taken along line 35-35;

(36) FIG. 36 is cross-sectional view of an iron-type golf club head including a lattice structure supporting a front face;

(37) FIG. 37 is a cross-sectional view of the iron-type golf club head of FIG. 31 taken along line 37-37;

(38) FIG. 38 is a cross-sectional view of the iron-type golf club head of FIG. 31 illustrating a slot formed in a rear side;

(39) FIG. 39 is an enlarged view of a portion of the cross-sectional view of the iron-type golf club head of FIG. 29;

(40) FIG. 40 is a cross-sectional view of the iron-type golf club head of FIG. 29 taken along line 40-40;

(41) FIG. 41 is a schematic illustration of a portion of a lattice structure according to the present disclosure;

(42) FIG. 42 is a schematic illustration of an intersection point of a lattice structure according to the present disclosure;

(43) FIG. 43 is a rear view of an iron-type golf club head having an external lattice structure according to the present disclosure;

(44) FIG. 44 is a cross-sectional view of the iron-type golf club head of FIG. 43 taken along line 44-44;

(45) FIG. 45 is a rear view of the iron-type golf club head of FIG. 43 with a filler material in the external lattice structure;

(46) FIG. 46 is a rear view of an iron-type golf club head illustrating a solid center of gravity plane according to the present disclosure;

(47) FIG. 47 is a rear view of the iron-type golf club head of FIG. 46 including a lattice structure;

(48) FIG. 48 is a top, back, right isometric view of the iron-type golf club head of FIG. 47;

(49) FIG. 49 is a right side view of the iron-type golf club head of FIG. 47;

- (50) FIG. 50 is a cross-sectional view of the iron-type golf club head of FIG. 47 taken along line 50-50;
- (51) FIG. 51 is a top view of the iron-type golf club head of FIG. 47;
- (52) FIG. 52 is a partial cross-sectional view of the iron-type golf club head of FIG. 51 taken along line 52-52;
- (53) FIG. 53 is a top view of the iron-type golf club head of FIG. 47 having a topline protrusion that covers the lattice structure along the topline;
- (54) FIG. 54 is a partial cross-sectional view of the iron-type golf club head of FIG. 53 taken along line 54-54;
- (55) FIG. 55 is a schematic illustration of a portion of a golf club head including a plug manufactured within a cavity;
- (56) FIG. 56 is a schematic illustration of the portion of the golf club head of FIG. 55 showing the movability of the plug;
- (57) FIG. 57 is a schematic illustration of a generally rectangular boundary manufactured around a powdered metal filler;
- (58) FIG. 58 is a schematic illustration of a generally round boundary manufactured around a powdered metal filler;
- (59) FIG. 59 is a schematic illustration of a generally oval or elliptical boundary manufactured around a powdered metal filler;
- (60) FIG. 60 is a rear view of a front face or striking face of a wood-type golf club head including a lattice structure;
- (61) FIG. 61 is a rear view of a front face or striking face of a wood-type golf club head including a ribbed structure;
- (62) FIG. 62 is a rear view of a front face or striking face wax mold for a wood-type golf club head including a ribbed structure;
- (63) FIG. 63 is a rear view of a front face or striking face for a wood-type golf club head formed from a casting made from the wax mold of FIG. 62;
- (64) FIG. 64 is a top, back, right isometric view of a iron-type golf club head wax mold according to the present disclosure;
- (65) FIG. 65 is a front view of a 3-D structure of the iron-type golf club head wax mold of FIG. 64;
- (66) FIG. 66 is a back view of a 3-D structure of the iron-type golf club head wax mold of FIG. 65
- (67) FIG. 67 is a top, front, left isometric view of a sintering support according to one aspect of the present disclosure;
- (68) FIG. 68 is a front view of the sintering support of FIG. 67;
- (69) FIG. 69 is a right side view of a sintering support according to another aspect of the present disclosure; and
- (70) FIG. 70 is a left side view of a sintering support according to the present disclosure.

DETAILED DESCRIPTION OF THE DRAWINGS

(71) The present disclosure is directed to golf club heads that are produced using an additive manufacturing process (i.e., printed layer by layer). In particular, a golf club head of the present disclosure includes a club head body that is manufactured using an additive manufacturing process and may be fabricated from a metal material or a metal alloy. In some embodiments, the club head body may include a segmented or lattice portion that is created during the additive manufacturing process and, therefore, is formed integrally with the club head body (i.e., the lattice portion and the club head body are a unitary component). In general, the incorporation of a segmented or lattice portion enables various material and/or performance characteristics of a golf club head to be selectively manipulated to achieve, for example a desired CG locations, MOI's, mass properties, face flex, distance variability, launch conditions, aesthetics, among other things.

(72) The use of the terms “segmented portion,” “lattice portion,” or “lattice structure,” herein refer to portions of a golf club head that are formed by one of a plurality of interconnected segments,

interconnected shapes, or connected surfaces. In some embodiments, the plurality of interconnected segments, interconnected shapes, or connected surfaces may be formed integrally with a club head body by an additive manufacturing process. In some embodiments, the lattice portion may define at least one cutout, or absence of material, that is formed within a unit cell (e.g., a repeated pattern defined by the lattice structure). The use of a lattice portion within a golf club head may allow various manufacturing and performance characteristics to be modified or customized. For example, a lattice portion may define a substantially reduced weight or density when compared to a solid material. As such, the placement of a lattice portion within a golf club head may be varied using an additive manufacturing process to selectively locate the CG of a golf club head in a desired location. In addition, the incorporation of a lattice portion into a golf club head may reduce the overall volume of material needed to manufacture the golf club head.

(73) The golf club heads disclosed herein may be manufactured using one or more of a variety of additive manufacturing processes. For example, a golf club head according to the present disclosure may be at least partially fabricated using a metal powder bed fusion additive manufacturing processes that fuses, melts, or bonds metal powder particles layer by layer along a build plane. In some embodiments, the metal powder particles may be melted or fused by a laser that forms cross-sections of a golf club head layer by layer along a build plane. In some embodiments, the metal powder particles may be melted or fused by an electron beam or ultrasonic energy to form cross-sections of a golf club head layer by layer along a build plane. In some embodiments, the metal powder particles may be bonded to form cross-sections of a golf club head layer by layer along a build plane via the deposit (e.g., printing) of a binder.

(74) The various methods of additive manufacturing used to manufacture a golf club heads according to the present disclosure may include binder jetting, direct energy deposition, selective laser melting (SLM), direct metal laser sintering (DMLS), fused deposition modeling (FDM), electron beam melting, laser powered bed fusion (LPBF), ultrasonic additive manufacturing, material extrusion, material jetting, Joule printing, electrochemical deposition, cold spray metal printing, DLP metal printing, Ultrasonic Consolidation or Ultrasonic Additive Manufacturing (UAM), LENS laser-based printing, electron beam freeform fabrication (EBF3), laser metal deposition, or carbon fiber additive manufacturing.

(75) Referring now to FIGS. 1-8, a putter-type club head **40** is shown in accordance with the present disclosure that may be formed through an additive manufacturing process. The club head **40** defines a body **42** and a face insert **44**, which may be coupled to one another after machining of the body **42**, as will be discussed in greater detail below. The body **42** defines a toe side **46**, a heel side **48**, a front side **50**, a top side or crown **52**, a bottom side or sole **54**, and a rear side **56**. Referring to FIG. 1, the body **42** of the club head **40** is formed from metallic and/or non-metallic materials. For example, the body **42** may be formed from any one of or a combination of aluminum, bronze, brass, copper, stainless steel, carbon steel, titanium, zinc, polymeric materials, and/or any other suitable material.

(76) The body **42** includes a front portion **60** and a rear portion **62**, the front portion **60** defining a face insert cavity **64** (see FIGS. 9 and 11), that is configured to receive the face insert **44**. The face insert **44** defines a striking surface **68**. The striking surface **68** comprises an entirety of the front surface of the face insert **44**, and is configured for contacting a golf ball. A peripheral edge **70** of the face insert **44** aligns with an inset edge **72** of the face cavity **64** (see FIG. 9) of the body **42**. The striking surface **68** further defines a first surface **74**, a second surface **76**, a third surface **78**, and a fourth surface **80** that define various angles with respect to a plane normal to the ground when the club head **40** is at address. The first surface **74** of the striking surface **68** may define an angle of about 1 degree, the second surface **76** may define an angle of about 2 degrees, the third surface **78** may define an angle of about 3 degrees, and the fourth surface **80** may define an angle of about 4 degrees. However, in some embodiments the surfaces **74**, **76**, **78**, **80** may define different angles, or may define the same angle. To that end, the striking surface **68** may comprise only a single, planar

surface that defines a constant angle.

(77) Referring to FIG. 3, the body **42** defines a toe portion or region **84**, a medial portion or region **86**, and a heel portion or region **88**. The heel region **88** of the body **42** includes a hosel **90** that extends upward therefrom. In some embodiments, the heel region **88** defines an aperture (not shown) that is disposed within the heel region **88**, which is configured to receive and secure a shaft (not shown) of the golf club (not shown).

(78) Referring specifically to FIGS. 3 and 4, the heel side **48** of the body **42** is rounded and extends from a lower heel-side inflection point **96** to an upper heel-side inflection point **98**. The sole **54** of the body **42** intersects with the heel side **48** at the lower heel-side inflection point **96**, while the crown **52** of the body **42** intersects with the heel side **48** at the upper heel-side inflection point **98**. The sole **54** defines a heel segment **100**, a medial segment **102**, and a toe segment **104**. The heel segment **100** and the toe segment **104** are generally angled and planar when viewed in elevation, while the medial segment **102** connects the toe segment **104** with the heel segment **100** and is generally planar. Further, portions of the medial segment **102** are parallel with respect to the ground (not shown) when the head **40** is at address.

(79) A portion of the toe segment **104** curves upward to a lower toe-side inflection point **108** where the toe segment **104** of the sole **54** intersects with the toe side **46**. A portion of the toe side **46** curves upward and inward, in a direction of the hosel **90**, and defines a generally straight portion of the toe side **46** that extends to an upper toe-side inflection point **110**. The top side **52** intersects with the toe side **46** at the upper toe-side inflection point **110**. When viewed from the front, the top side **52** extends laterally from the upper toe-side inflection point **110** to the upper heel-side inflection point **98**, and is interrupted by the hosel **90**.

(80) Referring to FIG. 3, the toe region **84**, the medial region **86**, and the heel region **88** are defined by vertical lines or planes P1 and P2 that extend through intersections of the heel segment **100** and the medial segment **102**, and the toe segment **104** and the medial segment **102**, respectively. The hosel **90** is located within the heel region **88**, and extends vertically from the top side **52**. In some embodiments, the hosel **90** may be at least partially disposed within the medial region **86**. The hosel **90** includes a plurality of cutouts **112** defined within a hosel arm **114**, which are generally in the shape of alternating triangles. The cutouts **112** may extend entirely through a width of the hosel **90**, or the cutouts **112** may not extend entirely through the hosel **90**, i.e., in the present embodiment, the hosel **90** does not include apertures that extend completely through the hosel arm **114**. In some embodiments, the cutouts **112** may align on a front and rear of the hosel **90** (see FIGS. 3 and 4). In alternative embodiments, only the front side of the hosel **90** may include the cutouts **112** or only the rear side of the hosel **90** may include the cutouts **112**. A shaft bore **116** extends from the hosel **90**, the shaft bore **116** being sized and shaped to receive a shaft (not shown), or an element that may be coupled with the shaft.

(81) Referring again to FIG. 1, a surface defining the front region **60** of the top side **52** is generally planar, while surfaces defining the rear region **62** of the top side **52** comprise a plurality of depressions, recesses, and other features. The front region **60** and the rear region **62** of the top side **52** are separated by a seam or groove **120** that extends from the heel side **48** to the toe side **46**. However, in embodiments that do not include the seam or groove **120**, the front region **60** and the rear region **62** are defined by a plane that extends vertically through the seam **120**. A shaft cavity **122** is further shown in FIG. 1, the shaft cavity **122** defining a cylindrical cavity within the shaft bore **116** into which the shaft (not shown) may be inserted. The shaft cavity **116** may be modified or formed to achieve any number of putter shaft positions, including heel, centered, and hosel offset.

(82) Still referring to FIG. 1, the face insert **44** is attached to or press fit within the insert cavity **64** of the body **42**. In some embodiments, the face insert **44** is secured and anchored via an interlocking structure (not shown). As provided in the cross-sectional views below, a bonding agent or adhesive **126** (see FIGS. 17 and 23) may be used to help secure the face insert **44** into the face

cavity **64**. Regardless of the type of retention mechanism used, the face insert **44** is fixed securely within the face cavity **64** of the body **42**.

(83) Referring now to FIG. 2, a rear view of the club head **40** is shown. A head cavity **130** is visible from the rear view, which houses a first weight **132**, a second weight **134**, and an internal lattice structure **136**. In some embodiments, the club head **40** may not include the first weight **132** and the second weight **134**. For example, the club head **40** may include solid material, the internal lattice structure **136**, or a cavity (i.e., no material) in place of the first weight **132** and the second weight **134**. In the illustrated embodiment, the lattice structure **136** is unitary with the body **42**, i.e., the lattice structure **136** comprises the same material and is manufactured at the same time as they body **42**. The first weight **132** and the second weight **134** are separate components, which may comprise tungsten or another type of metal. The lattice structure **136** is preferably 3D printed with the rest of the body **42**. However, in certain embodiments, the lattice structure **136** may comprise a separate cartridge that is insertable into the cavity **130**. The first weight **132** is located within the heel region **88**, the second weight **134** is located within the toe region **84**, and the lattice structure **136** extends across the heel region **88**, the toe region **84**, and the medial region **86**.

(84) Still referring to FIG. 2, the front portion **60** and the rear portion **62** of the club head **40** are shown separated by the groove **120**. As noted above, outer sides defining the front portion **60** are generally planar, while the rear portion **62** defines a rear upper side **140** and an inset region **142**. The inset region **142** defines a first or upper inset region **144** and a second or lower inset region **146**. The upper inset region **144** is defined by a first inset side **148**, which is a beveled edge that extends downward from the rear upper side **140** toward the sole **54**. A first inset platform **150** extends from portions of the first inset side **148**, the first inset platform **150** being generally parallel with respect to the rear upper side **140**. The first inset side **148** is generally U-shaped, and defines a periphery of the first inset region **144**.

(85) The second inset region **146** is also shown in FIG. 2, the second inset region **146** being defined by second inset sides **152** that are disposed on opposing sides of an alignment platform **154**. The alignment platform **154** includes a plurality of alignment notches or features **156**. The plurality of alignment features **156** may comprise any number and any type of designs that are sufficient to aid a golfer to align the putter-type golf club head **40** with a cup. In the present embodiment, the alignment features **156** are notches that are three dimensional features; however, in alternative embodiments, the alignment features **156** may be planar features, and may be painted along the alignment platform.

(86) A central alignment feature **158** is disposed centrally along the alignment platform **154**, and is configured to allow a golfer to align the putter head **40** with the cup before striking a golf ball (not shown). A window **160** is disposed within the second inset region **146**, between the second inset sides **152**, the window **160** being an opening that allows for air to enter the cavity **130** above the alignment platform **154**. As will be discussed in greater detail below, it is preferable to include blow through apertures along varying portions of a 3D printed putter head to allow excess material to be removed from the putter head **40** during the manufacture thereof, i.e., de-caking. It is for at least this reason that various apertures may be included along portions of the club **40** during at least some stages of the manufacturing process. Any commercial blower or air moving device may be used to blow excess material from within the putter head **40**.

(87) In some embodiments, a vacuum may be used to suck excess material from within the putter head **40**. In other embodiments, one or more tools including brushes, chisels, picks, or other implements are used to manually remove powder from within the putter head **40**. During post-printing processing, excess powder may be vacuumed or blown off of a build box that may include one or more of the putter heads **40**. After initially vacuuming or blowing, manual material removal is done to remove excess material from the putter head **40**. At this stage, remaining excess powder may be removed with one or more of the above-noted tools.

(88) Still referring to FIG. 2, the profiles of the alignment features **156** may define a variety of

shapes or cross sections that are sufficient to delineate the size and shape of the alignment features **156**. The alignment features **156** may define shallow grooves in the alignment platform **154**, the depths of which may be selected to sufficiently enable application and retention of a paint fill. In some embodiments, the alignment features **156** are filled with a paint or other organic coating that may be distinguished in appearance from its surrounding environment. In some embodiments, the grooves are partially or entirely filled with a material distinguished in appearance from its surrounding environment, e.g., a colored opaque or translucent polymer.

(89) Referring now to FIG. 4, the first and second weights **132**, **134**, the window **160**, the alignment platform **154**, the first inset side **148**, and the lattice structure **136** are shown in greater detail. The cutouts **112** along the hosel **90** are also visible in the rear view of FIG. 4. The first weight **132** and the second weight **134** are shown snugly disposed between an upper retention feature **166** and a lower retention feature **168**. The upper and lower retention features **166**, **168** generally define cylindrical portions having voids therebetween that allow the first and second weights **132**, **134** to be inserted therein, such that the first and second weights **132**, **134** fit snugly between the upper and lower retention features **166**, **168**. In some embodiments, a lock and key feature (not shown) within the cavity **130** retains the first and second weights **132**, **134** in place, so as to prevent undesired rotation of the first and second weights **132**, **134**.

(90) While the first and second weights **132**, **134** are shown having a particular diameter, varying types and sizes of weights are contemplated. In some embodiments, the weights **132**, **134** are removable, and may be removed and replaced by a user or a technician. As shown in the figures, the first and second weights **132**, **134** define an outer diameter D1 that is identical, and that is larger than an outer diameter D2 of the upper and lower retention features **166**, **168**. Further, while the first and second weights **132**, **134** are shown being disposed entirely within the heel region **88** and the toe region, respectively, it is contemplated that the first and second weights **132**, **134** may extend across one or more of the regions **100**, **102**, **104**.

(91) Still referring to FIG. 4, the lattice structure **136** is shown in greater detail. The lattice structure **136** is defined by a plurality of angled segments **172**, a plurality of horizontal segments **174**, and a plurality of vertical segments **176**, which combine to form a plurality of triangles or triangular portions. Air spaces are formed between the plurality of segments **172**, **174**, **176**, which may be filled with a filler material in some embodiments, as discussed in greater detail below. An outermost or rearmost row **178** (see FIG. 23) of the lattice structure **136** defines four separate right triangles, each of the right triangles being partially defined by one of the angled segments **172** and one of the vertical segments **176**. An innermost row **180** is also shown in FIG. 23. Curved rounds are defined at intersection points **182** of the segments **172**, **174**, **176**. The intersection points **182** are rounded (e.g., define a curvature, or a radius of curvature, and are not formed by the intersection of one or more straight lines) rather than cornered for manufacturing purposes. For example, it has been found that the overall strength of the lattice structure **136** is increased with the inclusion of curved rounds at the intersection points **182**. Through testing, it has been determined that when the intersection points **182** define sharp corners, the lattice structure is more likely to crack or break. Adding radii to sharp edges within geometry that is formed through 3D printing solves several issues, including: helping with de-caking (helps against green part destruction when blowing air against the lattice structure **136**), reducing sintering drag, and avoiding stress concentrations by adding radii on the edges of the lattice structure **136**.

(92) In some embodiments, the club head **40** may be 3D printed using binder jetting, which is a cost-effective way to produce low batch production with geometries that cannot be efficiently manufactured using conventional manufacturing methods. Metal binder jetting builds components by depositing (e.g., printing) a binding agent onto a layer of powder through one or more nozzles. The club head **40** is 3D printed, layer by layer, along of a first or build plane, as discussed in greater detail herein. The printing occurs at room temperature, or slightly above room temperature, which means that thermal effects are typically not present in the final printed components.

However, printing may occur at higher or lower temperatures. Metal binder jetting is a two-stage process, and involves a printing step and an essential post-processing step (sintering). Binder jetting involves spreading a thin layer of metal powder over a build platform, selectively depositing droplets of a binding agent that bonds the metal powder particles, and repeating the process until the build is complete. Once the build process is complete, the printed part may be excavated from the powder in the build platform and subsequently removed from the build platform. The result of the printing process is a part that is in the so-called “green” state, which is moved to a post-processing step to remove the binding agent and create the metal part.

(93) After the club head **40** has been printed, additional intermediate steps may be required before the club head **40** enters into a sintering step. In some embodiments, the part may need to go through a curing stage to allow the binder to set properly. Still further, in some embodiments before sintering, a de-binding step may be required to drive out any remaining binder. However, in some embodiments the curing step and the de-binding step may not be needed.

(94) There are two variations for the post-processing step. When using infiltration, the green part is first washed off from the binding agent to create a “brown” part with significant internal porosity, e.g., 70%. The brown part is then heated in an oven in the presence of a low-melting-point metal, such as bronze. The internal voids are filled, resulting in a bi-metallic part. When using sintering, the green part is placed in an industrial furnace. There, the binder is first burned off and the remaining metal particles are sintered together. The result is a fully metal component having dimensions that are approximately 20% smaller than the original green part. To compensate for shrinkage, the parts are printed larger, i.e., about 10%, or about 15%, or about 20%, or about 25%, or about 30% larger than final club head **40**. In some embodiments, the parts are printed between about 10% and about 30% larger, or between about 15% to about 25% larger, or between about 16% and about 20% larger. In some embodiments, the larger dimensions defined by the printed part (pre-sintering) may leave enough material to enable a printed club head to meet factory finish standards. In some embodiments, the golf club head may be machined (e.g., via milling or turning) post-sintering to obtain, for example, the loft, lie, weight, dimensions, volume, shape, etc., defined by the factory finish.

(95) In some embodiments, the club head **40** may be 3D printed using DMLS, or another one of the above-listed additive manufacturing techniques. In embodiments where the club head **40** is created using DMLS, a high powered laser is used to bond metal particles together, layer by layer, to create the club head **40**. While the process of DMLS involves fusing material particles to one another on a molecular level, many different metal alloys are compatible with this type of additive manufacturing technique. After printing, i.e., after a laser has selectively bonded the metal particles to one another, the club head **40** is cooled and loose powder is extracted. Post-processing steps may involve stress relief via thermal cycling, machining, heat treatment, or polishing. Various other post-processing steps may also be involved through printing of the club head **40** using DMLS or any of the above techniques.

(96) For example, in some additive manufacturing processes (e.g., DMLS) one or more supports (not shown) may be included on the club head **40** during printing to prevent the part from warping. Further, in DMLS, because the printed club head **40** is bonded to a build plate, a method of cutting may be required to cut the printed parts from the build plate. Electrical discharge machining (EDM) may be used to cut the printed parts from the build plate. Cutting or removing the parts may be required when using DMLS to build the parts, but may also be required when using other forms of additive manufacturing such as directed energy deposition DED or material extrusion.

(97) Referring now to FIGS. 5 and 6, side profiles of the club head **40** are shown in detail. More specifically, the toe side **46** is shown in FIG. 5, while the heel side **48** is shown in FIG. 6. The sole **54** or underside of the club head **40** is visible in the figures, and a plurality of design elements are visible spanning the front portion **60** and the rear portion **62** of the sole **54**. Fastener apertures **190** are also visible, the fastener apertures **190** being sized and shaped to allow fasteners **192** (see FIG.

8) to be inserted into the fastener apertures **190**, to thereafter retain the first and second weights **132**, **134** in position. The fastener apertures **190** are formed after the 3D printing process has occurred, i.e., in a post-printing state, as will be discussed in greater detail hereinafter below. Still referring to FIGS. **5** and **6**, the hosel **90** is shown in greater detail, the hosel **90** being disposed at an angle offset from a plane that is normal with respect to the ground. The front face **50** of the body **42** is also shown disposed at an offset angle with respect to a plane that is normal with respect to the ground when the club head **40** is at address. The front face **50** and the hosel **90** are angled in opposing directions with respect to the plane that is normal with respect to the ground when the club head **40** is at address.

(98) Referring now to FIGS. **7** and **8**, top and bottom views of the club head **40** are shown in detail. Referring specifically to FIG. **7**, the front portion **60** and the rear portion **62** are clearly shown being separated by the groove **120**. However, as noted above, in embodiments that do not include the groove **120**, the front portion **60** and the rear portion **62** are separated by a plane that extends through the groove **120**. The shaft bore **116** and shaft cavity **122** are also shown in greater detail. The shaft cavity **122** is disposed at an offset angle with respect to an axis normal to the ground when the club head **40** is at address. The planar portions along the front region **60** of the body **42** are also shown clearly in FIG. **7**. Further, the first and second inset regions **144**, **146** are depicted, and the plurality of alignment features **156** are shown surrounding the central alignment feature **158**. As illustrated in FIGS. **7** and **8**, a cutout region **196** is visible, the cutout region **196** following a profile of the second inset sides **152** and an outer edge **198** of the alignment platform **154** when viewed in the plan views of FIGS. **7** and **8**. While the term “cutout” is used herein, it should be appreciated that the manufacturing techniques utilized to create the club head **40** may or may not require the physical removal or grinding down of some portions, while certain portions do have to be removed or otherwise grinded down, as discussed with respect to FIGS. **9-13** below. As such, a “cutout” may refer to a portion that is devoid of material, not necessarily a portion that has had material that has been physically removed therefrom.

(99) Referring to FIG. **8**, a bottom view of the club head **40** is shown. Various design features **200** are shown spanning the front portion **60** and the rear portion **62** of the sole **54**, and two fasteners **192** are shown along the rear portion **62** of the sole **54**, the fasteners **192** being aligned with the weights **132**, **134**. The cutout region **196** is visible in FIG. **8**, which is shown defining various curved and straight surfaces. The fasteners **192** are shown disposed on opposing sides of the cutout region **196**. In some embodiments, the fasteners **192** are configured to be removed. However, in some embodiments, the fasteners **192** are permanently affixed to the club head **40** via an adhesive or another type of retention mechanism. The particular location of the fastener apertures **190** may be adjusted depending on a desired weight or center of gravity (CG) of the club head **40**. Still further, additional weights (not shown) may be added along the club head **40**. During manufacturing of the club head **40**, the weights **132**, **134** are inserted into the head cavity **130** and secured to the club head **40** via one or more fasteners, an adhesive, or another type of securement mechanism.

(100) Referring now to FIGS. **9-13**, a golf club head post-printed component **204** is shown. The post-printed component **204** depicts the club head **40** in a post-printed, pre grinded state. Further, the post-printed component **204** is shown without the face insert **44** applied to the body **42**, thus, the face insert cavity **64** is visible, the face insert cavity **64** being at least partially defined by an insert wall **206**. The post-printed component **204** is preferably formed using binder jetting, as described above. The post-printed component **204** may be printed at an angle that is offset by about 30 degrees with respect to the orientation shown in FIG. **12**, i.e., 30 degrees counterclockwise. In some embodiments, the post-printed component may be printed at an angle of between about 5 degrees and about 60 degrees offset, or between about 10 degrees and about 50 degrees offset, or between about 20 degrees and about 40 degrees offset from the orientation shown in FIG. **12**, i.e., from when the component **204** is at address.

(101) When manufacturing a golf club head via an additive manufacturing process, it is beneficial to ensure that the layer lines created during the additive manufacturing process avoid sharp surface interfaces (e.g., corners, edges, etc.) that fall along layer line edges. For example, in a binder jetting process, if a golf club head is printed such that the front face or striking surface is arranged parallel to the build plane (e.g., the front face is printed flat), the printed club head may show visible layer lines at shallow elevation changes, which may produce sharp corners that fall directly on a layer line edge and create cracks. The rotational offset that the post-printed component **204** is printed at, described above, may aid in preventing the printing of visible layer lines with sharp corners that fall on the layer line edge. In addition, printing at the rotation offset may prevent cracking of the green part during the print or sintering stages.

(102) Further, the rotational offset that the post-printed component **204** is printed at may also aid in Z-height limitations in, for example, a binder jetting process. For example, a thickness in the Z-direction (i.e., a height defined by a layer perpendicular to the build plane) may be reduced as the layers increase in Z-height during a binder jetting process. That is, the lower layers lay define an increased thickness relative to the upper layers due to weight of the overall structure weighing down on the lower layers. By printing the post-printed component at a rotational offset, the total Z-height defined by the component during the build is reduced, when compared to printing the component in the orientation of FIG. **12**.

(103) In some embodiments, the club head **40** may be printed in multiple components. For example, the hosel **90** and the body **42** may be printed, via binder jetting, as separate components. In this way, for example, the Z-height defined by the components being printed may be further reduced and the build efficiency (i.e., the amount of components printed during a build job) may be increased.

(104) Referring specifically to FIG. **9**, the face insert cavity **64** is shown in greater detail. The face insert cavity **64** is defined by the peripheral edge **70** that generally corresponds with an outer profile of the face insert **44** (see FIG. **1**) and the insert wall **206**. A material deposit **208** is centrally disposed within the face insert cavity **64** and extends outward from the insert wall **206**, the material deposit **208** defining a planar surface **210** and an outwardly extending platform. The material deposit **208** is intended to be machined off of the club head **40**. However, in some embodiments, only a portion of the material deposit **208** may be removed from the post-printed component **204**. The centrally disposed material deposit **208** may be provided or printed along what may be considered the “sweet spot” of the club head **40**. As a result, the machining of the centrally disposed material deposit **208** may allow for removal to enhance or otherwise modify the sweet spot. The location and size of any remaining portion of the material deposit **208** within the inset cavity may affect the characteristic time (“CT”) of the club head **40**.

(105) Still referring to FIG. **9**, a first or toe-side aperture **214** and a second or heel-side notch **216** are shown within the insert cavity **64**. As noted below, the heel-side notch **216** becomes the heel-side aperture **216** after processing of the post-printed component **204**. The toe-side aperture **214** is sized and shaped to allow air to flow through the post-printed component **204** during the manufacturing process to allow certain post-production material to be removed from the post-printed component **204**. The toe-side aperture **214** may also be sized and shaped to receive one or more portions of the face insert **44**, for example, in a lock-and-key fashion, so as to retain the face insert in place within the insert cavity **64**. During the post-processing step of manufacturing, the heel-side notch **216** is machined to become a heel-side aperture **216**, similar to the toe-side aperture. In some embodiments, there may be one or more additional apertures that are provided along the insert wall **206**. Since the particular face insert **44** described herein has regions defining different degrees, the face insert cavity **64** may be sized and shaped differently to receive alternatively shaped inserts.

(106) Referring now to FIG. **10**, the sole **54** of the post-printed component **204** is shown in greater detail. A rear side of the hosel **90** is also shown in greater detail. As shown in this particular view,

the post-printed component **204** includes several locations with additional material deposits **208** that are ultimately removed during a post-processing step. However, since the post-printed component **204** is depicted in a form after having been 3D printed, various portions of the post-printed component **204** include the material deposits **208**, which are machined off or are otherwise removed during a post-printing process. For example, and still referring to FIG. **10**, the material deposits **208** along the sole **54** that are cylindrical in nature are formed where the fastener apertures **190** are disposed in the final form of the club head **40**. Still further, one of the material deposits **208** is shown extending outwardly from the hosel **90**. The material deposits **208** may be formed in varying locations along the post-printed component **204**, which may exist after 3D printing because of one or more factors associated with 3D printing. For example, certain material deposits may be formed to enhance certain structural features of the post-printed component **204** during post-processing steps. Still further, material deposits may be formed or printed because of the technique that is utilized for manufacturing the post-printed component **204**, or to aid in verifying specifications, machining, polishing (as guides), or fixturing the post-printed component.

(107) In some embodiments, the one or more material deposits **208** may be provided so as to act as a reference circle to indicate a center of a desired bored or tapped hole. For example, the material deposits **208** located along the sole **54** are concentric circles that indicate where the hole should be drilled through in which the weights are located. The material deposits **208** may be a specified height so as to more easily machine portions of the post-printed component **204**.

(108) Referring to FIG. **11**, the material deposits **208** along the sole of the post-printed component **204** are shown more clearly. The toe-side aperture **214** is also shown in greater detail, and the upper and lower retention features **166**, **168** for the first weight **132** are visible through the toe-side aperture **214**. Guide holes **218** are shown disposed along the centrally raised planar surface **210**. A centered hole **219** is also shown in FIG. **11**, which, in combination with the guide holes **218**, are used to center the post-printed component **204** for various post-printing processes. For example, the centered hole **218** is located in the geometric center of the post-printed component **204**, and may be used as a machining “chuck” to elevate the post-printed component **204** and allow for machining of the various surfaces of the post-printed component. By placing the hole **219** centrally along the surface **210**, various efficiencies are achieved since it is preferable to elevate the post-printed component **204** by machining surfaces to tighter tolerances.

(109) Referring to FIG. **12**, the material deposit **208** that extends from the shaft bore **116** is shown in greater detail. Further, the shaft cavity **116** is entirely filled in, i.e., there is no shaft cavity **122** until the material disposed within the shaft bore **116** has been machined out to create the shaft cavity **122**. A rear view of the post-printed component **204** is shown in FIG. **13**, where the material deposits **208** are shown in greater detail. The heel-side aperture **216** is also visible through FIG. **13**, the heel-side aperture **216** being aligned with the upper and lower retention features **166**, **168** within the heel region **88**. The lattice structure **136** is visible in FIG. **13**, which is generally in the same configuration as it is within the club head **40**. While the foregoing description relating to the post-printed component **204** includes various aspects that are not shown or included within the club head **40**, alternative variations of the post-printed component **204** are contemplated that can achieve various aspects of the club head **40**. Once the post-printed component **204** is ready for sintering, the post-printed component is placed into a sintering furnace face down, i.e., with the face cavity **64** facing downward. In some embodiments, the post-printed component **204** may be sintered in an orientation other than face down. For example, the post-printed component **204** may be sintered sole down (i.e., with the sole **54** facing downward). Alternatively a sintering support (see FIGS. **67-70**) may be used to support the post-printed component **204** in a desired rotation orientation relative to gravity.

(110) Now turning to the views of FIGS. **14-17**, cross sectional views of the club head **40** are shown to illustrate the internal structure within the club head **40**. Referring specifically to FIG. **14**, some of the angled segments **172** of the lattice structure **136** are shown extending from an inner

surface **220** of the insert wall **206**. The heel-side aperture **216** is also shown, and a back side **222** of the face insert **44** is visible. The angled segments **172** of the lattice structure **136** extend from upper and lower ends of the insert wall **206**, toward the rear portion **62** of the club head **40**. A hosel bar **224** is also shown, the hosel bar **224** being aligned with the hosel **90**, but being disposed entirely within the cavity **130**. The hosel bar **224** is generally aligned with the hosel **90**, and extends vertically between the crown **52** and the sole **54** of the body **42**. One of the design elements **200** is further shown in FIG. **14** with a layer of the adhesive or bonding agent **126** disposed intermediate the design element **200** and the body **42**. Still further, circular protrusions **226** are shown extending outward from the inner surface **220** of the insert wall **206**. The circular protrusions **226** may be disposed along the inner surface **220** to aid with acoustics, altering the CT of the club head **40**, or for another reason.

(111) Referring now to FIG. **15**, another cross-sectional view of the club head **40** is shown. The angled segments **172** of the lattice structure **136** that extend from the inner surface **220** of the insert wall **206** are shown intersecting with other angled segments **172**. Referring specifically to the centrally located angled segments **172**, these angled segments **172** are offset from one another, such that the angled segments **172** are not disposed entirely within the same plane. Various other segments **172**, **174**, **176** are also offset from one another, such that intersecting segments **172**, **174**, **176** are not disposed within the same plane as one another. Still referring to FIG. **15**, the upper inset region **144** and the lower inset region **146** are partially shown, along with the various alignment features **156** along the alignment platform **154**. A portion of the central alignment feature **158** is also shown in FIG. **15**. The window **160** is further shown, with portions of the angled segments **172** being visible through the window **160**.

(112) Referring now to FIG. **16**, a cross-sectional view taken through the fasteners **192** and the weights **132**, **134** is shown. Various segments **172**, **174**, **176** are shown, which intersect at varying locations. Many of the intersection points **182** of the segments **172**, **174**, **176** are defined by the rounds, which may define acute, obtuse, or right angles. The fasteners **192** are further shown being disposed within the fastener apertures **190**, and retaining the weights **132**, **134** between the upper and lower retention features **166**, **168**. While the window **160** is shown being see-through, it is contemplated that an insert or another feature may be positioned within the window **160** to prevent debris from entering into the cavity **130** during use of the club head **40**. Still further, it is contemplated that a polymer or another type of filler material (not shown) may be disposed within the head cavity **130** such that the material is disposed within the lattice structure **136**. The material may be included to add weight or modify certain characteristics of the club head **40**. In some embodiments, the material may be added within the head cavity **130** to prevent materials such as dirt or other foreign matter from becoming engaged within the head cavity **130**.

(113) Referring to FIG. **17**, a cross-sectional view taken through the central alignment feature **158** is shown. The design feature **200** along the sole **54** of the club head **40** is shown with a layer of the adhesive **126** disposed between the design feature **200** and the body **42**. The face insert **44** is also shown with a layer of the adhesive **126** disposed between the face insert **44** and the insert wall **206**. One of the circular protrusions **226** is also shown extending from the inner surface **220** of the insert wall **206**. Varying other segments **172**, **174**, **176** of the lattice structure **136** are also shown extending across varying portions of the club head **40**. The first weight **132** is visible within the background of FIG. **17**. The upper inset region **144** and the lower inset region **146** are further shown, along with the upper inset edge **148** and the lower inset edge **152**.

(114) Now referring to FIGS. **18** and **19**, cross-sectional views of the club head **40** and the post-printed component **204** are shown, respectively, to illustrate contrasts between the club heads after and before post printing processing, respectively. The various material deposits **208** are visible in FIG. **19**, while the material deposits are shown having been removed, i.e., grinded down, drilled out, or otherwise machined in FIG. **18**. Further, the heel-side notch **216** in FIG. **19** has become the heel-side aperture **216** in FIG. **18**, which is achieved through drilling, grinding, or another type of

machining process. The shaft cavity **122** is also shown having been drilled out or otherwise machined in FIG. **18**, while the shaft cavity **122** is shown filled-in in FIG. **19**. Various other differences are visible between the pre- and post-processing versions of the club head, which may be achieved through a number of manufacturing techniques known to those skilled in the art. For example, certain surfaces and corners are grinded down or otherwise machined to achieve the club head **40** shown in FIG. **18**.

(115) Referring to FIG. **20**, a cross-sectional view of the club head **40** is shown that is taken through a center of the hosel **90**. The hosel notches **112**, which do not extend all the way through the hosel **90**, are shown, the hosel notches **112** taking various different forms along the hosel **90**. To that end, the hosel arm **214** is shown extending centrally through a center of the hosel **90**, the hosel arm **214** defining the various hosel notches **112** that are cutout from the hosel **90**. The hosel notches **112**, in some embodiments, may be disposed in a direction that is orthogonal with respect to the orientation shown in FIG. **20**. The groove **120** that separates the front region **60** and the rear region **62** of the body **42** is further shown in FIG. **20**, the groove **120** being generally v-shaped in cross-section.

(116) Referring now to FIGS. **21** and **22**, the face insert **44** is shown in greater detail. As noted above, the face insert **44** defines the striking surface **68**, which includes the first surface **74**, the second surface **76**, the third surface **78**, and the fourth surface **80**. In this particular embodiment, Descending Loft Technology™ is utilized, which comprises four flat surfaces that are milled into the face insert **44**. In a preferred embodiment, each of the surfaces **74**, **76**, **78**, **80** descends in loft by 1° from a top of the face insert **44** to a bottom of the face insert **44**. As a result of this configuration, when a player's shaft is pressed at impact, the ball contact will be higher on the face insert **44**. The face insert **44** therefore delivers consistent launch angles from putt to putt, which can lead to more consistent and predictable rolls.

(117) Referring now to FIG. **23**, a horizontal cross-sectional view of the club head **40** is shown. In this view, the hosel bar **224**, the first weight **132**, and the second weight **134** are shown in cross section. The fasteners **192** are also shown in cross section, along with the lattice structure **136**. The rear lattice row **182** and the front lattice row **180** are also shown. The front lattice row **180** and the rear lattice row **182** define a plurality of the segments **172**, **174**, **176**, which extend in a wide range of directions. In some embodiments, the disposition of the one or more segments **172**, **174**, **176** may be modified to change one or more characteristics of the club head **40**, such as the CG, CT, weight distribution, or another characteristic. Still further, in some embodiments, additional lattice rows may be added, and the segments **172**, **174**, **176** may be disposed in alternative configurations. As provided in FIG. **23**, the lattice structure **136** is generally limited to the medial region **86** of the club head **40**, with edge portions slightly crossing over into the heel region **88** and the toe region **84**. In some embodiments, the lattice structure **136** may extend entirely across one or more of the toe region **84**, the medial region **86**, and the heel region **88**. The lattice structure **136** may also extend only in a region defined between the first weight **132** and the second weight **134**.

(118) In general, the additive manufacturing principles and advantages of the putter-type club head **40** and the corresponding post-printed component **204** may be applied to other types of golf club heads. For example, an iron-type golf club head may be manufactured using an additive manufacturing technique and, in some embodiments, designed to include an internal or an external lattice structure or portion. The incorporation of a lattice structure into an iron-type golf club head via additive manufacturing may provide several manufacturing and performance advantages, in addition to enabling the design of an iron-type golf club head to leverage performance benefits from various iron club head designs.

(119) For example, conventional iron-type golf club heads may generally be designed with a muscle back design, a cavity back design, or a hollow construction. Typically, these conventional iron designs are limited in CG movement due to their volume and manufacturing method (e.g., forging, casting, metal injection molding, machined, etc.). Certain players may benefit from

playing a mid or large volume club head design that performs like a low volume club head. For example, hollow constructions are typically designed with a club face insert that may only be supported around a periphery of the face insert (e.g., the face insert is generally unsupported over the surface area that contacts a golf ball). Unsupported face inserts may provide inconsistent launch conditions and greater distance variability when compared to an iron design with a supported face (e.g., a muscle back design), but may provide greater distance and forgiveness. Additive manufacturing may allow for the design of a larger volume club head, which defines a higher MOI, with a supported face (e.g., similar to a low volume iron design) and the ability to adjust a CG location by adjusting mass and lattice structure locations.

(120) Referring now to FIGS. 24-27, an iron-type golf club head **300** is shown in accordance with the present disclosure that may be formed through an additive manufacturing process. The iron-type golf club head **300** includes a body **302** that defines an external skin or shell **304** that encloses an internal cavity **306** (see FIGS. 26 and 27). The external shell **304** may be formed around an external boundary of the body **302** (e.g., a boundary that is externally visible). In general, the iron-type golf club head **300** may be formed by an additive manufacturing process to define the appearance of a hollow construction iron design (e.g., a larger volume when compared to a muscle back design), which creates extra volume (i.e., the internal cavity **306**) within the external shell **304** to manipulate club head properties and/or performance. For example, the internal cavity **306** may be manipulated by adding solid material, a lattice structure, a weight, leaving it hollow, or any combination thereof to create unique CG locations and mass properties to influence face flex and performance.

(121) In general, the external shell **304** may form a thin border around a substantial portion or an entirety of the body **302** to give the appearance that the iron-type golf club head **300** is solid when viewed externally. The internal cavity **306** may be formed by a boundary defined by an inner periphery of the external shell **304**.

(122) The iron-type golf club head **300** defines a toe side **308**, a heel side **310**, a front side **312**, a top side **314**, a bottom side **316**, and a rear side **318**. The body **302** includes a toe region **320**, a medial region **322**, and a heel region **324**. Referring specifically to FIGS. 24 and 25, the toe region **320**, the medial region **322**, and the heel region **324** are defined by lines or planes P1 and P2 that extend through the iron-type golf club head **300** in a sole-topline direction **326** (e.g., a vertical direction from the perspective of FIGS. 24 and 25). The toe region **320** and the heel region **324** are arranged at laterally-opposing ends of the body **302**, and the medial region **322** is arranged laterally between the toe region **320** and the heel region **324**.

(123) The front side **312** of the body **302** may define a front face **327** that extends along the front side **312** of the body **302** from the toe region **320**, through the medial region **322**, and into at least a portion of the heel region **324**. In some embodiments, the front face **327** may define an entire front surface of the body **302** that extends laterally from the toe region **320**, through the medial region **322**, and into the heel region **324** to a junction between the front surface and a hosel **344** extending from the heel region **324**. In some embodiments, a portion of the front face **327** defined along the medial region **322** defines a striking face, which may include a plurality of laterally-extending grooves that are spaced from one another in the sole-topline direction **326** (see FIG. 39).

(124) The iron-type golf club head **300** defines a topline **328** extending laterally in a heel-toe direction **330** (e.g., a horizontal direction from the perspective of FIGS. 24 and 25) along the top side **314**, and a sole **332** extending laterally in the heel-toe direction **330** along the bottom side **316**. The toe region **320** includes a toe portion **334** of the body **302** that is defined by a portion of the body **302** between a distal end of the toe side **308** and the plane P1. In some embodiments, the plane P1 may be defined along a lateral edge of the grooves (not shown) formed in the front side **312** that is adjacent to the toe side **308**. In some embodiments, the plane P1 may intersect the top side **314** of the toe portion **334** at a toe-topline intersection point **336** along the topline **328** where the slope of a line tangent to the topline **328** is approximately zero (e.g., a point where a line

tangent to the periphery of the top side **314** is approximately parallel to the ground at address). In these embodiments, the plane P1 may extend through the toe portion **334** in the sole-topline direction **326** to a toe-sole intersection point **337**.

(125) The heel region **324** includes a heel portion **338** of the body **302** that is defined by a portion of the body **302** between a distal end of the heel side **310** and the plane P2. In some embodiments, the plane P2 may be defined along a lateral edge of the grooves (not shown) formed in the front side **312** that is adjacent to the heel side **310**. In some embodiments, the plane P2 may intersect the top side **314** at a heel-topline inflection point **340** (e.g., a point where the periphery of the top side **314** transitions from concave down to concave up). In these embodiments, the plane P2 may extend through the heel portion **338** in the sole-topline direction **326** to a heel-sole intersection point **342**.

(126) The heel portion **338** includes the hosel **344** that extends from the heel portion **338** at an angle (e.g., a lie angle formed between a plane parallel to the ground on which the club head rests at address and a center axis defined through the hosel **344**) in a direction away from the toe portion **334**. The hosel **344** defines a hosel cavity **346** (see FIG. 26) within which a shaft (not shown) may be inserted for coupling to the iron-type golf club head **300**. In some embodiments, a ferrule (not shown) may abut or be at least partially inserted into the hosel **344**. In some embodiments the hosel cavity **346** may extend through at least a portion of the hosel **344**.

(127) The topline **328** may extend along an outer periphery of the top side **314** of the body **302** from the heel-topline inflection point **340**, along the medial region **322**, to the toe-topline intersection point **336**. The sole **332** may extend along a periphery of the bottom side **316** of the body **302** from the toe-sole intersection point **337**, along the medial region **322**, to the heel-sole intersection point **342**.

(128) With reference to FIGS. 26-31, the internal cavity **306** of the body **302** includes a lattice structure **348** arranged within at least a portion of the internal cavity **306**. For example, in some embodiments, the lattice structure **348** may extend in the sole-topline direction **326** along the entire internal cavity **306** (see FIGS. 26 and 27). In some embodiments, the lattice structure **348** may extend in the sole-topline direction **326** along a portion of the internal cavity **306**. For example, the lattice structure **348** may extend from an end of the internal cavity **306** adjacent to the topline **328** to a location between the topline **328** and the sole **332** (see FIGS. 28 and 29). Alternatively, the lattice structure **348** may extend from an end of the internal cavity **306** adjacent to the sole **332** to a location between the sole **332** and the topline **328** (see FIGS. 30 and 31).

(129) In some embodiments, the lattice structure **348** may extend laterally in the heel-toe direction **330** along substantially the entire internal cavity **306**. For example, the lattice structure **348** may extend laterally in the heel-toe direction **330** from the toe region **320**, through the medial region **322**, and into at least a portion of the heel region **324**. In some embodiments, the lattice structure **348** may extend laterally in the heel-toe direction **330** a distance defined by a lateral extension of the front face **327** (e.g., the lattice structure **348** may extend the same lateral distance as the front face **327**).

(130) In general, the incorporation of the lattice structure **348** into the internal cavity **306** defines a lower density relative to a solid material (e.g., solid metal) filling within the internal cavity **306** of the same volume. Since the lattice structure **348** defines a lower density compared to a solid material (e.g., solid metal) filling of the same volume, a CG volume ratio defined as a ratio between a volume $V_{sub.L}$ that the lattice structure **348** occupies in the internal cavity **306** to a volume $V_{sub.S}$ that a solid portion **349** occupies within the internal cavity **306** may be altered to move the CG location in the sole-topline direction **326**. In other words, an orientation of the lattice structure **348** between the topline **328** and the sole **332** (e.g., a distance that the lattice structure **348** extends over the internal cavity **306** in the sole-topline direction **326**) and the volume ratio may define a CG defined by the body **302**. The orientation of the lattice structure **348** and the volume ratio may be altered to define a desired CG location for the iron-type golf club head **300**.

(131) With specific reference to FIGS. 28-31, the arrangement, dimensions, and volume of the

lattice structure **348** within the internal cavity **306** may be customized to define a high CG (e.g., a CG arranged closer to the topline **328**) or a low CG (e.g., a CG arranged closer to the sole **332**). For example, the iron-type golf club head **300** illustrated in FIGS. **28** and **29** may define a CG point **350** that is higher (e.g., closer to the topline **328**) when compared to a CG point **352** defined by the iron-type golf club head **300** illustrated in FIGS. **30** and **31**. This is due to the differences in the arrangement, dimensions, and volume of the lattice structure **348** illustrated in FIGS. **28-31**. For example, arranging the lattice structure **348** adjacent to the topline **328** (see FIGS. **28** and **29**) and filling a reminder of the internal cavity **306** adjacent to the sole **332** with the solid portion **349** (e.g., solid metal material that is formed layer by layer) provides more high density material adjacent to the sole **332** and, thereby, lowers the CG of the iron-type golf club head **300**. Conversely, arranging the lattice structure **348** adjacent to the sole **332** (see FIGS. **30** and **31**) and filling a reminder of the internal cavity **306** adjacent to the topline **328** with the solid portion **349** provides more high density material adjacent to the topline **328** and, thereby, raises the CG of the iron-type golf club head **300**.

(132) The incorporation of the lattice structure **348** in the internal cavity **306** of the iron-type golf club head **300** enables the CG location to be manipulated to any location between a CG defined by a completely solid body (e.g., the internal cavity **306** is completely filled with solid material) and a CG defined by a completely hollow body (e.g., the internal cavity **306** is completely hollow or devoid of material). It should be appreciated that the volumes defined by the lattice structure **348** (V.sub.L) and the solid portion **349** (V.sub.S) of the internal cavity **306** do not need to be discretely defined along the sole-topline direction **326**. That is, in some embodiments, the lattice structure **348** may include one or more solid portions **349** arranged on vertically-opposing sides thereof. For example, the lattice structure **348** may not originate from an internal side of the external shell **304** adjacent to the top side **314** or an internal side of the external shell **304** adjacent to the bottom side **316**. Rather, the internal cavity **306** may include solid portions **349** that extend from the top and bottom internal sides of the external shell **304** that form the internal cavity **306** and the lattice structure **348** may be arranged between the solid portions **349**. Alternatively, the internal cavity **306** may include one or more lattice structure **348** that are separated in the sole-topline direction **326** with the solid portion **349** arranged therebetween.

(133) In some embodiments, the variability and control over the CG location provided by the incorporation of the lattice structure **348** into the iron-type golf club head **300** may be leveraged when designing and manufacturing a set of iron-type golf club heads. For example, a set of irons may include long irons (e.g., 1-iron through 5-iron), mid irons (e.g., 6-iron through 9-iron), short irons (e.g., pitching wedge through lob wedge), and it may be desirable to define varying CG locations for each iron within a set. In some embodiments, the various types of irons within a set may define varying CG locations (e.g., long irons define a low CG, mid irons define a middle CG, and short irons define a high CG, or another configuration). In any case, a set of iron-type golf club heads according to the present disclosure may include at least two iron-type golf club heads manufactured via an additive manufacturing process with a lattice structure incorporated in both of the iron-type golf club heads at varying CG volume ratios to define different CG locations along the sole-topline direction **326** for each of the iron-type golf club heads produced.

(134) In some embodiments, a set of iron-type golf club heads according to the present disclosure may include a first golf club head and a second golf club head. The first golf club head may define a first orientation of a first lattice structure between a sole and a topline and a first volume ratio between a first lattice volume and a first solid volume. The second golf club head may define a second orientation of a second lattice structure between a sole and a topline and a second volume ratio between a second lattice volume and a second solid volume. In some embodiments, the second orientation may be different than the first orientation to define a different CG between the first golf club head and the second golf club head. In some embodiments, the second volume ratio may be different than the first volume ratio to define a different CG between the first golf club head

and the second golf club head. In some embodiments, the second orientation may be different than the first orientation and the second volume ratio may be different than the first volume ratio to define a different CG between the first golf club head and the second golf club head.

(135) In addition to the ability of the lattice structure **348** to manipulate the CG location of the iron-type golf club head **300**, a stiffness defined along the external shell **304** in the regions occupied by the lattice structure **348** may be maintained, for example, similar to the stiffness support provided by the solid portion **349**. For example, in some embodiments, the front face **327** of the body **302** is supported by (e.g., in engagement with) one of the lattice structure **348** and the solid portion **349** along an entire surface area thereof, which prevents local areas of non-uniform or reduced stiffness. In this way, for example, the incorporation of the lattice structure **348** into the iron-type golf club head **300** enables the iron-type golf club head **300** to provide the advantages of various iron designs to a user. For example, the iron-type golf club head **300** may provide the consistent launch conditions and distance variability of a low volume (e.g., muscle back) iron design and the increased MOI of a mid or high volume iron design.

(136) In some embodiments, the iron-type golf club head **300** may be designed to provide enhanced distance (e.g., a utility iron) and may include a lattice structure that is attached to the body but does not support the front face or a face insert coupled to the body. For example, with reference to FIGS. **32** and **33**, the iron-type golf club head **300** may include a face insert **354** that is coupled to the front side **312** of the body **302** and attached (e.g., via welding) around a periphery of the front side **312**. When the face insert **354** is coupled to the body **302**, the internal cavity **306** may be enclosed by the external shell **304** and the face insert **354**, and the lattice structure **348** may be enclosed within the internal cavity **306**.

(137) In the illustrated embodiment, the lattice structure **348** extends from the internal surfaces of the external shell **304**. For example, the lattice structure **348** may be attached to or supported by the internal surfaces of the external shell **304** on the body **302** adjacent to the toe side **308**, the heel side **310**, the top side **314**, the bottom side **316**, and the rear side **318**. The lattice structure **348** may be interrupted by the solid portion **349** that, in the illustrated embodiment, extends along the bottom side **316** from the toe region **320** to a location between the toe region **320** and the heel region **324** (see FIG. **32**). When the face insert **354** is attached to the body **302**, a gap **356** may be formed between a termination plane T defined by the lattice structure **348** (e.g., a plane generally parallel to the face insert **354** along which the lattice structure **348** terminates) and the face insert **354**. In other words, the lattice structure **348** may be set back from the face insert **354** leaving the face insert **354** unsupported by the lattice structure **348**.

(138) The design and construction of the iron-type golf club head **300** illustrated in FIGS. **32** and **33** provides support for the face insert **354** only around the periphery thereof and creates a stiffer body structure, which allows the face insert **354** to be thinner, thereby enhancing performance (e.g., increased distance). In some embodiments, the face insert **354** may be manufactured via an additive manufacturing process. As described herein, in some embodiments, an orientation of a golf club head relative to a build plane during an additive manufacturing process may improve the quality and performance of the green part of the final post-sintering product. In configurations where a face insert is not planar and includes, for example, a portion of a sole integrated with a striking surface (i.e., an L-cup face insert), it may be beneficial to orient the face insert such that the front face or striking surface is rotationally offset from the build plane. In this way, for example, the layer lines formed during the additive manufacturing process may not pass through the edge where the front face transitions to the sole (e.g., a leading edge). That is, if the face insert were manufactured with the front face oriented parallel to the build plane, a layer line may pass through the leading edge of the face insert, which may cause defects in the green part and/or the post-sintered part. This is avoided by printing the face insert with the front face rotationally offset relative to the build plane. In addition, the rotational orientation of the face insert relative to the build plane may be tailored to maximize efficiency of the additive manufacturing process (i.e., arrange as many face inserts

within a given build area to manufacture as many face inserts as possible during a build).

(139) With reference to FIGS. **34** and **35**, in some embodiments, an aperture **358** may be formed, for example, via additive manufacturing, that extends laterally into and through the solid portion **349**. The aperture **358** may extend laterally from the toe side **308** of the solid portion **349** to a location between the toe side **308** and an end of the solid portion **349**. In some embodiments, the aperture **358** may be filled with a weight bar **360** (e.g., tungsten). The incorporation of the weight bar **360** may aid in lowering the CG location of the iron-type golf club head **300** along the sole-topline direction **326**.

(140) In some embodiments, the iron-type golf club head **300** may be manufactured to enable the weight bar **360** to be attached or secured within the aperture **358** via a sintering process. For example, the weight bar **360** may be manufactured with dimensions that are a predetermined percentage larger than the factory finish dimensions. In some embodiments, the predetermined percentage may be about 10%, or about 15%, or about 20%, or about 25%, or about 30% larger than the factory finish dimensions of the iron-type golf club head. In some embodiments, the predetermined percentage may be between about 10% and about 30%, or between about 15% to about 25%, or between about 16% and about 20%. In some embodiments, the weight bar **360** may be manufactured via an additive manufacturing process. In some embodiments, the weight bar **360** may be formed by a metal injection molding process. In any case, once the weight bar **360** is initially manufactured with dimensions that are the predetermined percentage larger than the factory finish dimensions, the weight bar **360** may go through a sintering process. During the sintering process, the weight bar **360** may shrink to at least one of the factory finish dimensions. For example, the weight bar **360** may shrink to a factory finish diameter, but may still define a length that is longer than a factory finish length to enable the weight bar **360** to be cut to length and conform to the outer profile of the body **302** during post-processing.

(141) Similar to the weight bar **360**, the body **302** of the iron-type golf club head **300** may be manufactured with dimensions that are a predetermined body percentage larger than the factory finish dimensions. In some embodiments, the body **302** may be manufacturing via a binder jetting process and the predetermined body percentage may be about 10%, or about 15%, or about 20%, or about 25%, or about 30% larger than the factory finish dimensions of the iron-type golf club head. In some embodiments, the predetermined percentage may be between about 10% and about 30%, or between about 15% to about 25%, or between about 16% and about 20%. Once the body **302** is initially manufactured with dimensions that are larger than the factory finish dimensions, the body **302** may go through a sintering process. Prior to the sintering process, the post-sintered weight bar **360** may be inserted into the body **302** at the predefined location (e.g., the aperture **358**). The sintering process may shrink the body **302** to the factor finish dimensions. During the sintering process, the body **302** may shrink around the weight bar **360** and form an interference fit between the body **302** and the weight bar **360**, thereby securing the weight bar **360** within the body **302** without requiring any secondary adhesion techniques (e.g., welding, adhesive, etc.).

(142) By first sintering the weight bar **360** and then sintering the body **302** with the post-sintered weight bar **360** installed within the body **302**, the iron-type golf club head **300** may naturally form an interference fit between the body **302** and the weight bar **360**, which secures the weight bar **360** within the body **302**. For example, once the weight bar **360** is sintered, it may be substantially prevented from further shrinkage, which allows the secondary sintering of the iron-type golf club head **300** to shrink around the weight bar **360** and form a natural interference fit therebetween. In addition, this manufacturing process avoids issues that may arise due to sintering a golf club head that includes metals with different densities. For example, if the weight bar **360** and the body **302** were sintered for the first time together, the weight bar **360** would shrink more than the body **302** due to increased density relative to the body **302**. As such, the weight bar **360** may not fit within the body **302** post-sintering and add inefficiencies to the manufacture of the iron-type golf club head **300**. Further, the staged sintering process avoids issues that arise due to different metals requiring

different sintering temperatures. In general, this staged sintering process may be used to couple a weight bar to a body of a golf club head as long as the weight bar and a cavity within which the weight bar is to be arranged define a similar or the same shape.

(143) In some embodiments, rather than a weight insert, the density of a golf club head according to the present disclosure may be controlled by the additive manufacturing process. For example, in a DMLS process, a speed at which the laser translates over a component and creates a layer is proportional to a density of the metal formed. As such, a speed at which the laser translates over selective portions when manufacturing a golf club head layer by layer may be controlled to define a desired density profile over the entire volume of the golf club head. In the embodiment of FIGS. 34 and 35, the laser may be slowed down when traversing over portions of the iron-type golf club head 300 within the solid portion 349. In this way, for example, the solid portion 349 may include at least a portion thereof that defines a higher density and aids in lowering the CG of the iron-type golf club head 300, similar to the weight bar 360.

(144) In some embodiments, the iron-type golf club head 300 may be designed to incorporate a lattice structure that is attached only behind the front face and does not support a remainder of the body. With reference to FIG. 36, the iron-type golf club head 300 may include the lattice structure 348 arranged on a rear surface of the front face 327. A thickness (e.g., a distance that the lattice structure 348 extends away from the front face 327 along a direction parallel to a normal defined by the front face 327) defined by the lattice structure 348 may be dimensioned such that lattice structure 348 only engages the front face 327 and the remainder of the body 302 may be unsupported by the lattice structure 348. In other words, a gap 362 may be arranged between the lattice structure 348 and a rear portion 364 of the body 302 (e.g., a portion of the body 302 arranged rearward of the front face 327), which may be fabricated from solid material (e.g., solid metal that is formed layer by layer), along an entire area defined by the front face 327. In this way, for example, the stiffness of the front face 327 may be increased (e.g., when compared to a front face/face insert without a lattice structure connected thereto). In some embodiments, for example, the lattice structure 348 may be arranged over a portion of the rear surface of the front face 327, rather than an entirety of the rear surface.

(145) The increased stiffness provided by the lattice structure 348 being attached to the front face 327 may provide more consistent launch conditions and improved distance variability similar to a low volume (e.g., muscle back) iron design. In addition, a shape, size, and mass distribution in the rear portion 364 may be easily tailored or customized via additive manufacturing to allow for variations in CG location, MOI, etc.

(146) As described herein, the size, shape, volume, and arrangement of the lattice structure 348 within the body 302 of the iron-type golf club head 300 may be controlled or designed to provide stiffness to selective portions of the body 302, the front face 327, and/or the face insert 354. With the lattice structure 348 acting as a local stiffening structure, the location of the lattice structure 348 within the body 302 may directly impact performance of the iron-type golf club head 300 (e.g., sound, feel, ball speed, distance variability, launch conditions, etc.).

(147) In some embodiments, the stiffness differences in the front face 327 provided by the support or lack thereof by the lattice structure 348 may be leveraged to produce a set of iron-type golf club heads with varying face stiffness. Similar to conventional iron-type golf club sets that transition from cavity back/hollow construction to muscle back design as they transition from long irons to short irons, the design of the iron-type golf club head 300 may be varied using additive manufacturing to provide varying performance characteristics as the iron-type golf club heads transition from long irons to short irons. For example, a set of iron-type golf club heads according to the present disclosure may include at least two iron-type golf club heads that transition from a front face or face insert that is not supported by a lattice structure to a front face or face insert that is at least partially supported by a lattice structure to leverage the performance benefits of these different designs described herein in a single set of iron-type golf club heads.

(148) As described herein, there are several performance and design advantages to incorporating a lattice structure into an iron-type golf club head, or another type of golf club head, via additive manufacturing. In order to effectively manufacture the iron-type golf club head according to the present disclosure certain design aspects should be considered. For example, many additive manufacturing processes utilize a metal powder bed to produce components layer by layer, as described herein. Similar to the putter-type golf club head **40**, iron-type golf club heads may be required to be de-caked of residual metal powder that remains after the initial scavenging of the printed component from the powder bed.

(149) In general, an iron-type golf club head according to the present disclosure may define a flow path that extends through the body to allow a fluid (e.g., gas) to be forced through or sucked out of the body. In some embodiments, the flow path may be formed via apertures or slots formed in the body and may extend through a lattice structure. Referring to FIGS. **37** and **38**, in some embodiments, the body **302** may define a flow path **366** that extends along the internal cavity **306** and the hosel cavity **346**. Specifically, the lattice structure **348** may be formed by a plurality of segments that form a plurality of cutouts, or absences of material, between the plurality of segments. In this way, for example, fluid flow may occur through the lattice structure **348**. In some embodiments, the lattice structure **348** may include shapes or surfaces that define one or more cutouts, or absences of material, to enable fluid flow therethrough.

(150) The internal cavity **306**, including the lattice structure **348** formed therein, may be in fluid communication with the hosel cavity **346** and at least one other aperture or slot formed in the body **302**. For example, with specific reference to FIG. **38**, a slot **368** may be formed in the rear side **318** of the body **302** that extends laterally across the body **302**. In this embodiment, the flow path **366** may extend from the hosel cavity **346**, along the lattice structure **348**, and through the slot **368** to define a flow path that extends through the body **302**. In this way, for example, pressurized fluid (e.g., gas), a vacuum, a brush, a tool, or gravity may be applied to the flow path **366** to aid in removing powdered metal and excess material from the additive manufacturing process (i.e., de-caking).

(151) In some embodiments, the iron-type golf club head **300** may not include the slot **368** and, rather, may include an aperture (not shown in FIG. **37**, but see, e.g., aperture **358** in FIG. **35** and externally accessible and/or visible lattice structures in FIGS. **43**, **44**, **45**, **47**, and **49**) formed, for example, in the toe portion **334**. The aperture (not shown in FIG. **37**) formed in the toe portion **334** may extend into the internal cavity **306** to provide fluid communication with the lattice structure **348**. The aperture (not shown in FIG. **37**) may be utilized after manufacturing the body **302** via an additive manufacturing process to provide compressed fluid (e.g., gas) or a vacuum to the flow path **366** to aid in removing powdered metal and excess material. After the leveraging the flow path **366** for the de-caking process, the aperture may be plugged, for example, by a screw or a plug to prevent debris from entering the internal cavity **306** during use.

(152) In some applications, the arrangement and number of openings that form a flow path may be varied dependent on the type additive manufacturing process being used to form a golf club head. For example, in an additive manufacturing process where the manufactured part defines a density that is close to a solid metal part (e.g., SLM, DMLS, etc.), the number of openings in a flow path may be reduced when compared to an additive manufacturing process where the manufactured part defines a lower density and higher porosity (e.g., binder jetting). The lower density and high porosity defined by the green part after a binder jetting process may be susceptible to damage if high pressure fluid is used to remove excess metal powder from the part. For example, blowing the metal powder over the green part after a binder jetting process may act like a sand blaster and affect the quality of the green part. For these reasons, it may be beneficial to include at least two openings in a flow path for a golf club head manufactured using, for example, a binder jetting process. In any case, a golf club head manufactured using an additive manufacturing process may be designed to include at least one opening into a flow path from which excess material may be removed from the

manufactured part.

(153) As described herein, an iron-type golf club head according to the present disclosure may be manufactured using a binder jetting process, an SLM, a DMLS additive manufacturing process, or another direct laser metal melting process. In DMLS, for example, support structures are leveraged to attach the component being manufactured to a build plate and to protect against warping/distortion that may occur due to the high temperatures utilized during the additive manufacturing process. In some instances, when a lattice structure is created by an additive manufacturing process (e.g., DMLS), it may need support structures during printing. It is advantageous to avoid creating support structures because they are difficult to remove, especially from internal cavities and overhangs. The necessity for support structures is dependent on the additive manufacturing process, orientation of the lattice structure, and design of the lattice within the club head.

(154) In some embodiments, a golf club head manufactured using an additive manufacturing process according to the present disclosure may include a lattice structure that is self-supporting and does not require internal supports to be created. In general, print orientation (i.e., the orientation of a build plane along which the golf club head is formed layer by layer) relative to lattice structure design can ensure that the lattice structure is self-supporting.

(155) Referring to FIGS. **39** and **40**, a second plane or build plane B may be defined as a plane along which the iron-type golf club head **300** is printed layer by layer during the additive manufacturing process. In the illustrated embodiment, the build plane B is rotationally offset from a first plane or ground plane G defined by the body **302** and that is arranged parallel to the ground on which the iron-type golf club head **300** may be placed at address. As described herein, when manufacturing a golf club head via an additive manufacturing process, it is beneficial to ensure that the layer lines created during the additive manufacturing process avoid sharp surface interfaces (e.g., corners, edges, etc.) that fall along layer line edges. To leverage the benefits of avoiding sharp surface interfaces that fall along layer line edges, the build plane B may be rotationally offset from the ground plane G, when viewed from the toe side **308** (see FIG. **39**) or the heel side **310**, which results in the iron-type golf club head **300** being printed layer by layer at an angle that is offset by about 30 degrees with respect to the ground plane G (e.g., 30 degrees clockwise from the perspective of FIG. **39**). In some embodiments, the iron-type golf club head **300** may be printed along a build plane B that is offset at an angle of between about 0 degrees and about 175 degrees, or between about 5 degrees and 160 degrees, or between about 5 and about 140 degrees, or between about 5 degrees and 120 degrees, or between about 5 degrees and 90 degrees, or 5 degrees and about 60 degrees, or between about 10 degrees and about 50 degrees, or between about 20 degrees and about 40 degrees with respect to the ground plane G.

(156) The lattice structure **348** may define one or more lattice build angles relative to the build plane B. Each of the lattice build angles is defined along a common plane defined by the lattice structure **348**. For example, the lattice structure **348** may be formed by a plurality of segments **370** that extend from an internal boundary of the internal cavity **306** to either another internal boundary or the solid portion **349**. In the illustrated embodiment, the internal cavity **306** may be formed by an internal sole surface **372**, an internal rear surface **374**, an internal front surface **376**, and an internal top surface **378**. The internal top surface **378** is formed by the interface between the solid portion **349** and the lattice structure **348**.

(157) In the illustrated embodiment, the lattice structure **348** defines a plurality of planes along which the plurality of segments **370** extend. With specific reference to FIG. **39**, the lattice structure **348** defines a lattice plane L1 that forms a lattice build angle A1 with respect to the build plane B, and a lattice plane L2 that forms a lattice build angle A2 with respect to the build plane B. Each of the lattice planes L1, L2 is formed by a portion of the plurality of segments **370** that are aligned and oriented at the respective lattice build angle A1, A2 relative to the build plane B. The lattice structure **348** includes a plurality of portions that align with the lattice planes L1, L2, which are

spaced from one another a distance that is governed by the length and orientation of the plurality of segments **370** within the lattice structure **348**.

(158) With specific reference to FIG. **40**, the lattice structure **348** defines a lattice plane L3 that forms a lattice build angle A3 with respect to the build plane B, and a lattice plane L4 that forms a lattice build angle A4 with respect to the build plane B. Each of the lattice planes L2, L3 is formed by a portion of the plurality of segments **370** that are aligned and oriented at the respective lattice build angle A3, A4 relative to the build plane B. The lattice structure **348** includes a plurality of portions that align with the lattice planes L3, L4, which are spaced from one another a distance that is governed by the length and orientation of the plurality of segments **370** within the lattice structure **348**. In general, the sequential spacing and intersection between each of the lattice planes L1, L2, L3, L4 forms the geometry of the lattice structure **348**.

(159) Through testing, it has been determined that when the build plane B is oriented parallel to a normal extending from the front face **327**, the lattice structure **348** is self-supporting with lattice build angles A1, A2, A3, A4 that are each greater than or equal to 30 degrees. That is, if each of the lattice build angles A1, A2, A3, A4 is greater than or equal to 30 degrees, the lattice structure **348** may be additively manufactured without any additional support structures, for example, during DMLS. In this way, for example, the need to remove support structures on the lattice structure **348** during the post-processing stages may not be required, which significantly improves manufacturing efficiency, costs, and time.

(160) In the illustrated embodiment, each of the lattice planes L1, L2, L3, L4 extend in varying directions and form a plurality of intersection points **380** where one or more of the plurality of segments **370** that form the lattice planes L1, L2, L3, L4 intersect. In the illustrated embodiment, each of the intersection points **380** may be formed by the intersection of six of the segments **370** extending from the intersection point **380** in a different direction (see FIG. **41**), except at locations where the intersection point **380** is interrupted by one or more of the internal sole surface **372**, the internal rear surface **374**, the internal front surface **376**, and the internal top surface **378** (or another surface in engagement with the lattice structure **348**), or a termination plane along which the lattice structure **348** terminates prior to engaging a surface (see FIG. **33**).

(161) With specific reference to FIG. **41**, in one embodiment, the lattice structure **348** may define a unit cell **382** that is formed by a cutout, air space, or absence of material defined between interconnected intersection points **380** that occur along a common plane. For example, in the illustrated embodiment, the lattice structure **348** may define square-, rectangular-, or diamond-shaped unit cells **382**. This geometry defined by the unit cells **382** may be similar to the lattice structure **348** illustrated in FIGS. **26-40**. However, a lattice structure according to the present disclosure is not limited to this shape of unit cell and alternative geometries may be utilized. For example, as described herein, the segments **172**, **174**, **176** of the lattice structure **136** define generally triangular-shaped cutouts or air spaces. Alternatively or additionally, in some embodiments, at least a portion of the unit cells in a lattice structure according to the present disclosure may define a pentagonal shape, a hexagonal shape, or any other polygonal shape.

(162) In some embodiments, a unit cell defined by a lattice structure according to the present disclosure can be formed by interconnected shapes (e.g., ovals, circles, or another geometric shape) with varying orientation to form a repeated pattern, or unit cell. In some embodiments, a lattice structure according to the present disclosure may be formed by a differential geometry structure. For example, a lattice structure according to the present disclosure may be formed by a gyroid structure that includes a plurality of interconnected, periodic minimal surfaces. The gyroid structure may define a unit cell that is repeated in a pattern over a desired volume to form a lattice structure according to the present disclosure. In general, the use of a differential geometry structure (e.g., a gyroid) may reduce stress concentrations formed along the lattice structure due to the reduction in sharp edges formed on the lattice structure, which may provide similar advantages as adding curvature, described herein with reference to the lattice structure **136**. In some embodiments, a

lattice structure according to the present disclosure may define a tublane structure or a plate-lattice structure.

(163) Regardless of the design and properties of the lattice structure, a golf club head according to the present disclosure may be manufactured via additive manufacturing to include a lattice structure formed integrally with at least a portion of a body, a front face, and/or a face insert of the golf club head. During manufacture, when the build plane is oriented parallel to the front face normal, each portion of the lattice structure may be printed at an angle greater than or equal 30 degrees relative to the build plane to ensure that the lattice structure is self-supporting and does not require support structures.

(164) In some embodiments, a lattice structure according to the present disclosure may define a hybrid or variable structure that varies in one or more of unit cell type, unit cell geometry, unit cell size, segment length, segment, thickness, segment volume, and unit cell density at one or more locations along the lattice structure. For example, in embodiments of the iron-type golf club head **300** where the lattice structure **348** is connected to the front face **327**, the lattice structure **348** may be varied behind the front face **327** to improve or maximize ball speed over the front face **327**, more specific to where players actually impact the golf ball (e.g., lower (closer to the sole) than a face center point). For example, the lattice structure **348** may vary in a thickness, size, and/or shape of the segments **370**, a density of the unit cells **382**, and/or a shape or type of the unit cells **382** at various locations behind the front face **327**.

(165) As described herein, adding curvature or removing sharp edges within geometry that is formed through additive manufacturing solves several issues, including: helping with de-caking (e.g., helps against green part destruction when blowing air against a lattice structure), reducing sintering drag, and avoiding stress concentrations in a lattice structure. In the embodiments where a lattice structure according to the present disclosure is formed via a plurality of segments, the intersection points may be curved at each intersection between the segments at the intersection point. For example, FIG. **42** illustrates an embodiment of an intersection point **380** taken along a cross-sectional plane. As illustrated in FIG. **42**, the lattice structure **348** may define rounded edges at each intersection between the segments **370** forming the intersection point **380**. That is, the intersecting edge formed between each of the intersecting segments **370** may be rounded to define a curvature or a radius of curvature, rather than culminating at a point. In addition to the intersection points **380**, each edge of the segments **370** formed in the lattice structure **348** may define a rounded or curved edge.

(166) In general, a lattice structure according to the present disclosure may define rounded or curved edges along, for example, edges of intersection points, edges of segments forming the lattice structure, and any other edges formed along the lattice structure to provide the manufacturing and performance benefits described herein.

(167) In the embodiments of FIGS. **26-40**, the lattice structure **348** is arranged internally with respect to the body **302** (e.g., at least partially within the internal cavity **306**). In some embodiments, a golf club head may be designed to include an externally accessible/visible lattice structure. For example, a golf club head according to the present disclosure may include at least one external face that is formed at least partially by a lattice structure. As described herein, removing residual metal powder may be required following the manufacture of a golf club head via an additive manufacturing process. In some embodiments, a golf club head according to the present disclosure may include apertures and/or define a flow path to enable the removal of excess metal powder. Another solution to aiding in removal of metal powder from a 3D printed golf club head may be to arrange the lattice structure such that it is externally accessible/visible. In some embodiments, a depth that an externally-facing lattice structure extends into a body of the golf club head and/or a unit cell size (e.g., volume or surface area) of the lattice structure may be limited to ensure efficient de-caking of residual metal powder present after manufacturing the golf club head via an additive manufacturing process.

(168) Referring to FIGS. 43 and 44, an iron-type golf club head 400 is shown in accordance with the present disclosure that may be formed through an additive manufacturing process. The iron-type golf club head 400 includes a body 402 and an externally-facing lattice structure 404 formed on the body 402. In general, the lattice structure 404 may be formed along a portion of an externally-facing face or surface of the body 402 in place of solid material, which reduces a weight of the iron-type golf club head 400 and maintains stiffness (e.g., similar to the stiffness provided by solid material).

(169) The iron-type golf club head 400 defines a toe side 408, a heel side 410, a front side 412, a top side 414, a bottom side 416, and a rear side 418. The body 402 includes a toe region 420, a medial region 422, and a heel region 424. The toe region 420, the medial region 422, and the heel region 424 are defined by lines or planes P1 and P2 that extend through the iron-type golf club head 400 in a sole-topline direction 426 (e.g., a vertical direction from the perspective of FIG. 43). The toe region 420 and the heel region 424 are arranged at laterally-opposing ends of the body 402, and the medial region 422 is arranged laterally between the toe region 420 and the heel region 424.

(170) The front side 412 of the body 402 may define a front face 427 that extends along the front side 412 of the body 402 from the toe region 420, through the medial region 422, and into to at least a portion of the heel region 424. In some embodiments, the front face 427 may define an entire front surface of the body 402 that extends laterally from the toe region 420, through the medial region 422, and into the heel region 424 to a junction between the front surface and a hosel 444 extending from the heel region 424. In some embodiments, a portion of the front face 427 defined along the medial region 422 defines a striking face, which may include a plurality of laterally-extending grooves (not shown) that are spaced from one another in the sole-topline direction 426.

(171) The iron-type golf club head 400 defines a topline 428 extending laterally in a heel-toe direction 430 (e.g., a horizontal direction from the perspective of FIG. 43) along the top side 414, and a sole 432 extending laterally in the heel-toe direction 430 along the bottom side 416. The toe region 420 includes a toe portion 434 of the body 402 that is defined by a portion of the body 402 between a distal end of the toe side 408 and the plane P1. In some embodiments, the plane P1 may be defined along a lateral edge of the grooves (not shown) formed in the front side 412 that is adjacent to the toe side 408. In some embodiments, the plane P1 may intersect the top side 414 of the toe portion 434 at a toe-topline intersection point 436 along the topline 428 where the slope of a line tangent to the topline 428 is approximately zero (e.g., a point where a line tangent to the periphery of the top side 414 is approximately parallel to the ground at address). In these embodiments, the plane P1 may extend through the toe portion 434 in the sole-topline direction 426 to a toe-sole intersection point 437.

(172) The heel region 424 includes a heel portion 438 of the body 402 that is defined by a portion of the body 402 between a distal end of the heel side 410 and the plane P2. In some embodiments, the plane P2 may be defined along a lateral edge of the grooves (not shown) formed in the front side 412 that is adjacent to the heel side 410. In some embodiments, the plane P2 may intersect the top side 414 at a heel-topline inflection point 440 (e.g., a point where the periphery of the top side 414 transitions from concave down to concave up). In these embodiments, the plane P2 may extend through the heel portion 438 in the sole-topline direction 426 to a heel-sole intersection point 442.

(173) The heel portion 438 includes the hosel 444 that extends from the heel portion 438 at an angle (e.g., a lie angle formed between a plane parallel to the ground on which the club head rests at address and a center axis defined through the hosel 444) in a direction away from the toe portion 434. The hosel 444 defines a hosel cavity (not shown) within which a shaft (not shown) may be inserted for coupling to the iron-type golf club head 400. In some embodiments, a ferrule (not shown) may abut or be at least partially inserted into the hosel 444.

(174) The topline 428 may extend along an outer periphery of the top side 414 from the heel-topline inflection point 440, along the medial region 422, to the toe-topline intersection point 436.

The sole **432** may extend along a periphery of the bottom side **416** from the toe-sole intersection point **437**, along the medial region **422**, to the heel-sole intersection point **442**.

(175) The lattice structure **404** of the iron-type golf club head **400** may be designed and manufactured with similar properties and characteristics as the lattice structures disclosed herein. In the illustrated embodiment, the lattice structure **404** may define at least a portion of a rear face **446** of the body **402**. The rear face **446** may extend over at least a portion of the rear side **418** of the iron-type golf club head **400**. For example, the lattice structure **404** may extend laterally (e.g., in the heel-toe direction **430**) over the medial region **422** and at least a portion of each of the toe region **420** and the heel region **424**. The lattice structure **404** may extend along the sole-topline direction **426** between a rear-topline edge **448** and a rear-sole edge **450**.

(176) Referring specifically to FIG. **44**, the lattice structure **404** may define an external border **452** of the body **402** along the rear face **446**. In some embodiments, the external border **452** may define an externally-facing border of the lattice structure **404** (e.g., a border of the lattice structure **404** that is externally visible/accessible). The lattice structure **404** may define a thickness **454** that the lattice structure **404** extends into the body **402**, for example, in a direction arranged generally normal to a rear surface **455** defined by the body **402**. In some embodiments, the thickness **454** may be about 5 millimeters. In some embodiments, the thickness **454** may be between about 4 millimeters and about 6 millimeters, or between about 3 millimeters and about 7 millimeters. In some embodiments, the thickness **454** may be less than or equal to about 5 millimeters.

(177) In some embodiments, the thickness **454** defined by the lattice structure **404**, in combination with the lattice structure **404** defining the external border **452** of the body **402**, may enable the lattice structure **404** to be easily de-caked after printing of the iron-type golf club head **400**. In the illustrated embodiment, the lattice structure **404** may include unit cells that define a generally triangular shape. In some embodiments, the lattice structure **404** may define unit cells of any shape or design according to the present disclosure. In some embodiments, a size and shape of the unit cells defined by the lattice structure **404** also be customized to ensure that the de-caking process occurs efficiently.

(178) Referring to FIG. **45**, after the iron-type golf club head **400** is manufactured via an additive manufacturing process, the excess metal powder may be easily removed from the externally-accessible lattice structure **404**, and the lattice structure **404** may be filled with a filler material **456**. In some embodiments, the filler material **456** may be a light weight (e.g., low density) epoxy or resin. In some embodiments, the filler material **456** may be substantially transparent or translucent. Filling the lattice structure **404** with the filler material **456** efficiently prevents debris from collecting in the lattice structure **404** and, in some embodiments, may maintain the external visibility of at least the external border **452** of the lattice structure **404**.

(179) As described herein, incorporating a lattice structure into a golf club head provides several manufacturing, performance, and customizable advantages. In some embodiments, a lattice structure may be utilized to efficiently distribute the mass throughout a golf club head. For example, in conventional golf club heads, solid material present above a horizontal plane (e.g., a plane that extends in the heel-toe direction) defined by the CG is inefficient, since it limits movement of the CG. In some embodiments, a golf club head according to the present disclosure may replace the solid material rearward of the front face and above a CG plane defined by the golf club head with a lattice structure. In this way, the stiffness provided by the solid material may be maintained by the lattice structure, and the replacement of the solid material with the lattice structure reduces a density in the replaced areas, which allows the saved mass to be used elsewhere on the golf club head to improve performance.

(180) Referring to FIG. **46**, an iron-type golf club head **500** is shown in accordance with the present disclosure. The iron-type golf club head **500** may define a cavity back design and may be fabricated from solid material (e.g., solid metal). The iron-type golf club head **500** defines a solid CG plane C (i.e., a plane that extends in a heel-toe direction that aligns with a CG defined by a

solid configuration of the iron-type golf club head **500**) that extends laterally across a body **502** of the iron-type golf club head **500**. According to embodiments of the present invention, the solid material arranged rearward of a front face (e.g., a striking face) and above (e.g., upward from the perspective of FIG. **46**) the solid CG plane C on the body **502** of the iron-type golf club head **500** may be replaced by a lattice structure (see FIG. **47**). In some embodiments, the solid CG plane C may be defined as a plane extending parallel to the ground plane G at a location defined by the CG of the body **502** when the body **502** is fabricated from solid material.

(181) Referring now to FIGS. **47-50**, the iron-type golf club head **500** may include an externally accessible/visible lattice structure **504**. The iron-type golf club head **500** defines a toe side **508**, a heel side **510**, a front side **512**, a top side **514**, a bottom side **516**, and a rear side **518**. The body **502** includes a toe region **520**, a medial region **522**, and a heel region **524**. The toe region **520**, the medial region **522**, and the heel region **524** are defined by lines or planes P1 and P2 that extend through the iron-type golf club head **500** in a sole-topline direction **526** (e.g., a vertical direction from the perspective of FIG. **47**). The toe region **520** and the heel region **524** are arranged at laterally-opposing ends of the body **502**, and the medial region **522** is arranged laterally between the toe region **520** and the heel region **524**.

(182) The front side **512** of the body may define a front face **527** that extends along the front side **512** of the body **502** from the toe region **520**, through the medial region **522**, and into to at least a portion of the heel region **524**. In some embodiments, the front face **527** may define an entire front surface of the body **502** that extends laterally from the toe region **520**, through the medial region **522**, and into the heel region **524** to a junction between the front surface and a hosel **544** extending from the heel region **524**. In some embodiments, a portion of the front face **527** defined along the medial region **522** defines a striking face, which may include a plurality of laterally-extending grooves (not shown) that are spaced from one another in the sole-topline direction **526** (see FIG. **50**).

(183) The iron-type golf club head **500** defines a topline **528** extending laterally in a heel-toe direction **530** (e.g., a horizontal direction from the perspective of FIG. **47**) along the top side **514**, and a sole **532** extending laterally in the heel-toe direction **530** along the bottom side **516**. The toe region **520** includes a toe portion **534** of the body **502** that is defined by a portion of the body **502** between a distal end of the toe side **508** and the plane P1. In some embodiments, the plane P1 may be defined along a lateral edge of the grooves (not shown) formed in the front side **512** that is adjacent to the toe side **508**. In some embodiments, the plane P1 may intersect the top side **514** of the toe portion **534** at a toe-topline intersection point **536** along the topline **528** where the slope of a line tangent to the topline **528** is approximately zero (e.g., a point where a line tangent to the periphery of the top side **514** is approximately parallel to the ground at address). In these embodiments, the plane P1 may extend through the toe portion **534** in the sole-topline direction **526** to a toe-sole intersection point **537**.

(184) The heel region **524** includes a heel portion **538** of the body **502** that is defined by a portion of the body **502** between a distal end of the heel side **510** and the plane P2. In some embodiments, the plane P2 may be defined along a lateral edge of the grooves (not shown) formed in the front side **512** that is adjacent to the heel side **510**. In some embodiments, the plane P2 may intersect the top side **514** at a heel-topline inflection point **540** (e.g., a point where the periphery of the top side **514** transitions from concave down to concave up). In these embodiments, the plane P2 may extend through the heel portion **538** in the sole-topline direction **526** to a heel-sole intersection point **542**.

(185) The heel portion **538** includes the hosel **544** that extends from the heel portion **538** at an angle (e.g., a lie angle formed between a plane parallel to the ground on which the club head rests at address and a center axis defined through the hosel **544**) in a direction away from the toe portion **534**. The hosel **544** defines a hosel cavity (not shown) within which a shaft (not shown) may be inserted for coupling to the iron-type golf club head **500**. In some embodiments, a ferrule (not shown) may abut or be at least partially inserted into the hosel **544**.

(186) The topline 528 may extend along an outer periphery of the top side 514 from the heel-topline inflection point 540, along the medial region 522, to the toe-topline intersection point 536. The sole 532 may extend along a periphery of the bottom side 516 from the toe-sole intersection point 537, along the medial region 522, to the heel-sole intersection point 542.

(187) With specific reference to FIGS. 47-49, the lattice structure 504 of the iron-type golf club head 500 may be designed and manufactured with similar properties and characteristics as the lattice structures disclosed herein. In the illustrated embodiment, the lattice structure 504 may include unit cells that define a generally triangular shape. In some embodiments, the lattice structure 504 may define unit cells of any shape or design according to the present disclosure.

(188) The lattice structure 504 extends over a portion of the body 502 that is arranged above the solid CG plane C (e.g., in a direction from the sole 532 toward the topline 528) and rearward (e.g., in a direction from the front side 512 toward the rear side 518, or to the left from the perspective of FIG. 49) of the front face 527. For example, the rear surface 546 of the front face 527 may extend along a plane R at an angle relative to the sole-topline direction 526, which is defined by the loft of the iron-type golf club head 500. The plane R along which the rear surface 546 extends may intersect with the solid CG plane C and the plane R and the solid CG plane C may define the boundaries of the lattice structure 504.

(189) By replacing solid material with the lattice structure 504, the density defined by the body 502 in these regions may be locally reduced and the stiffness previously provided by the solid material may be maintained. In this way, for example, the CG of the iron-type golf club head 500 may be lowered (e.g., moved in a direction toward the sole 532) compared to a golf club head made from solid material (i.e., relative to the solid CG plane C). For example, a CG volume ratio defined as a ratio between a volume $V_{sub.L}$ that the lattice structure 504 occupies to a volume $V_{sub.S}$ that solid material occupies may be a factor in defining a CG location in the sole-topline direction 526.

(190) In addition, the mass removed by the lattice structure 504 may be redistributed to other locations on the body 502 to improve performance, as desired. For example, if a mass of a golf club head is maintained and the solid material above a solid CG plane and rearward of the front face is replaced by a lattice structure, the reduced density provided by the lattice structure may enable mass to be redistributed to other regions of the golf club. In some embodiments, it may be desirable to lower a CG defined by a factory finished golf club head, when compared to a solid-material golf club head. In this embodiment, the mass saved by incorporating the lattice structure may be redistributed toward the sole of the golf club head. Redistributing this weight may further lower the CG of the golf club head and this process may repeat until the CG and the redistribution of the saved mass replaced by a lattice structure converge. That is, the golf club head may continue to be replaced with lattice structure in design, until the amount of volume replaced by a lattice structure and the redistributed mass converge on a CG location. Thus, the replacement of the solid material in a golf club head may be an iterative process in design and the final finish product may be produced with a CG that balances volume replaced by lattice structure and redistributed mass.

(191) Referring to FIGS. 51 and 52, in some embodiments, the lattice structure 504 may be externally visible and form at least a portion of the topline 528. In some instances, a golfer may not wish to view the lattice structure 504 at address (i.e., along the topline 528). Referring to FIGS. 53 and 54, in some embodiments, the topline 528 may be formed by solid material, for example, by a topline protrusion 548 that extends laterally along the topline 528.

(192) As described herein, weight distribution (e.g., CG manipulation) in a golf club head may be manipulated via additive manufacturing processes. In some embodiments according to the present disclosure, a golf club head may be manufactured layer by layer to include a cavity within a generally solid portion of a golf club head. During manufacture, the cavity may be filled with a plug or weight that is not permanently bound or attached to the internal surfaces of the cavity. As such, the plug or weight may be held in place by the surrounding metal powder in the cavity but not attached to the surfaces that form the cavity. That is, the plug or weight may be arranged free-

floatingly within the cavity. In this way, for example, once the metal powder is removed from the cavity, a position of the plug or weight within the cavity may be manipulated to distribute the weight of the plug at a desired location within the cavity. For example, an orientation of the golf club head may be manipulated and gravity may be used to alter a position of the plug or weight within the cavity. The position of the plug or weight within the cavity may be secured, for example, by filling the cavity with a filler material (e.g., a plastic resin, a foam material, etc.). In general, the plug or weight arranged within the cavity may generally define any shape or structure that defines a weight that may be manipulated to alter a weight distribution within the golf club head. In some embodiments, the plug or weight may be fabricated from the same material as the surrounding solid portion of the golf club head. In some embodiments, the plug or weight may be fabricated from a material that is different than a material that is used to fabricate the surrounding solid portion of the golf club head. In some embodiments, the plug or weight may be fabricated from a material with a higher density than a material that is used to fabricate the surrounding solid portion of the golf club head. In some embodiments, the plug or weight may be fabricated from a material with a lower density than a material that is used to fabricate the surrounding solid portion of the golf club head. (193) FIG. 55 illustrates one embodiment of a portion of a golf club head **600** that includes a cavity **602** within which a plug or weight **604** is formed during an additive manufacturing process. In some embodiments, the portion of the golf club head **600** is a portion of a body that is desired to be formed of solid material. During additive manufacture of the portion of the golf club head **600**, the layer by layer forming of the portion of the golf club head **600** enables the formation of the cavity **602** and the plug **604** within the cavity **602**. The plug **604** may be manufactured such that the plug **604** is spaced from the internal surfaces that form the cavity **602**. In the illustrated embodiment, the plug **604** is surrounded by residual metal powder **606**. The metal powder **606** may hold the plug **604** in place within the cavity **602**, while maintaining the detachment between the plug **604** and the internal surfaces of the cavity **602**.

(194) One or more ports **608** may be in communication with the cavity **602** to enable the removal of the metal powder **606** after the portion of the golf club head **600** is manufactured. In the illustrated embodiment, the portion of the golf club head **600** includes two ports **608** arranged at opposing sides of the cavity **602**. In some embodiments, the portion of the golf club head **600** may include more or less than two ports **608** arranged in any orientation that connects to the cavity **602**.

(195) After the portion of the golf club head **600** is manufactured, pressurized fluid (e.g., gas), a vacuum, a brush, a tool, or gravity may be applied to the one or more ports **608** to remove the excess metal powder **606** surrounding the plug **604**. As illustrated in FIG. 56, once the metal powder **606** is removed, the plug **604** may be free to move within the cavity **602**. In this way, for example, a position of the plug **604** may be manipulated to alter a weight distribution within the portion of the golf club head **600**. In the illustrated embodiment, the cavity **602** and the weight or plug **604** defines a generally cylindrical shape. In some embodiments, the cavity **602** and the weight or plug **604** may define any shape (e.g., rectangular, polygonal, or any other 3-D shape) as required by the shape and structure defined by the portion of the golf club head **600** within which the cavity **602** is arranged. For example, in some embodiments, the cavity **602** and the plug or weight **604** may define similar shapes. In some embodiments, the weight or plug **604** may define a different shape than the cavity **602** as long as the weight or plug **604** is capable of displacing within the cavity **602**, once the excess material is removed from the cavity **602**.

(196) In the illustrated embodiment, the design and shape of the cavity **602** and the plug **604** enable the weight distribution to be moved in a lateral direction (e.g., left and right from the perspective of FIG. 56). In some embodiments, the design and shape of the cavity **602** and the plug **604** may be altered to enable the weight distribution within the portion of the golf club head **600** to be moved in any direction, as desired. For example, the cavity **602** and the plug **604** may be designed to allow for the weight distribution to be moved in a heel-toe direction, a sole-to-pline direction, an oblique direction, and/or between a front face and a rear face within the portion of the golf club head **600**.

Moving the weight distribution, via movement of the plug **604** to a desired location may alter the performance characteristics of a golf club head, for example, by moving the CG and/or placing more weight in a heel or a toe of the golf club head.

(197) Once the plug **604** is positioned in a location within the cavity **602** according to a desired weight distribution, the cavity **602** may be filled with a low-density filler material to secure the position of the plug **604** within the cavity **602**. For example, the low-density filler material may be a plastic material, a resin material, and/or a foam material.

(198) In some embodiments, binder material may be selectively added around solid portions of a golf club head to form a border or shell surrounded by metal powder. Then, during the sintering post-processing stage the metal powder enclosed within the border may solidify forming the appropriate solid portion of a golf club head. In this way, for example, use of a binder during a binder jetting process may be reduced while printing golf club heads, thereby improving manufacturing costs and efficiency.

(199) As described herein, at least a portion of a golf club head that is manufactured using an additive manufacturing process may include a solid portion (e.g., a volume region that is intended to be solid metal in the factory finish part). In some embodiments, an additive manufacturing process according to the present disclosure may improve efficiency and quality of the manufactured part by forming a boundary that includes at least one layer around a portion of a golf club head and post-processing the portion of the golf club head form the portion of the golf club head within the boundary as a solid portion.

(200) For example, as described herein, the material deposit **208** formed on the post-printed component **204** and the solid portion **349** on the iron-type golf club head **300** may be formed from solid material (e.g., solid metal). In some embodiments, these solid material portions on a golf club head may be formed using an additive manufacturing process by printing a boundary that includes at least one layer of printed material and surrounds a volume of unprinted material (e.g., metal powder). For example, the material deposit **208** or the solid portion **349** may be formed by printing a boundary that encloses a volume and is formed by at least one layer during an additive manufacturing process. The volume enclosed by the boundary may be filled with powdered metal and, thereby, may be constrained (i.e., cannot move) within the volume. The manufactured golf club head may then be sintered, which transitions the powdered metal enclosed within the volume to solid material (e.g., solid metal). By only requiring at least one layer of material to form a solid volume on a golf club head, the amount of time, binder material (e.g., for a binder jetting process), and/or power (e.g., for a SLM or a DMLS process) may be reduced, which may provide reduced costs and increase efficiency during the additive manufacturing process.

(201) FIGS. **57-59** illustrate embodiments of a cross-section of a solid volume of a golf club head that is manufactured during an additive manufacturing process. In the illustrated embodiments, a solid volume **700** includes a boundary **702**. In some embodiments, the boundary **702** may be formed by at least one layer that is created during an additive manufacturing process. In some embodiments, the boundary **702** may be formed by at least two, at least three, at least four, or five or more layers during an additive manufacturing process. The boundary **702** may enclose the solid volume **700** and the powdered metal arranged within the solid volume **700**. The powdered metal enclosed by the boundary **702** may be maintained or supported by the boundary **702** (i.e., prevented from displacing after the additive manufacturing process), once the boundary **702** is fully formed. With the boundary **702** fully formed, the powdered metal enclosed therein may be formed into solid metal via a sintering process. In some embodiments, forming solid metal portions in a golf club head via sintering powdered metal enclosed by a boundary may produce higher densities when compared to solid metal portions that are formed completely layer by layer. In this way, for example, the cost and efficiency of the additive manufacturing process may be improved for creating a golf club head and the quality of the manufactured part may be improved.

(202) As illustrated in FIGS. **57-59**, the cross-sectional shapes of the solid volumes **700** may take

various shapes and sizes. In the illustrated embodiments, the boundary **702** formed around the solid volume **700** may be a rectangular, a round, or an oval shape. In some embodiments, the boundary **702** and/or the solid volume **700** may take any shape or size that is required by the desired factory finish golf club head. For example, any solid portion of a golf club head may be enclosed with a boundary that takes any shape, and the golf club head may be sintered to transition the volume enclosed by the boundary into solid material.

(203) As described herein, additive manufacturing provides several design, manufacturing, and performance benefits for golf club heads. Additive manufacturing also provides several advantages to the development or prototyping of golf club heads. For example, an entire set of iron-type golf club heads may be printed within a single build platform (e.g., a powdered metal bed used in binder jetting, DMLS, SLM, etc.). As such, an entire set of iron-type golf club heads may be printed and tested in a single build job, which differs, for example, from a forging process where the golf club heads are formed one at a time. Alternatively or additionally, multiple iterations of a golf club head design may be printed and tested during a single build job.

(204) As described herein, in some embodiments according to the present disclosure, at least a portion of a golf club head may be manufactured via an additive manufacturing process. In some embodiments, a golf club head may be at least partially manufactured, or at least partially formed via a mold that is manufactured, via an additive manufacturing process. For example, a face insert that defines a striking face or a front face of a golf club head may be designed to include a 3-D structure that improves performance. In some embodiments, a rear side of a front face on a golf club head may include a lattice structure or a ribbed structure.

(205) For example, as illustrated in FIG. **60**, a face insert **800** defines a front face or striking face of a wood-type golf club head and may include a lattice structure **802** arranged on a rear side of the front face or striking face. In the illustrated embodiment, the lattice structure **802** includes generally triangularly-shaped unit cells that vary in density, surface area, or volume along the rear side of the face insert **800**. In some embodiments, the lattice structure **802** may define any size or shape according to the lattice structures described herein. In any case, the lattice structure **802** may vary in one or more of unit cell type, unit cell geometry, unit cell size, segment length, segment, thickness, segment volume, and unit cell density at one or more locations along the rear side of the face insert **800**. In some embodiments, the variability in the lattice structure **802** along the rear side of the face insert **800** may provide improved performance, when compared to a lattice structure with constant properties.

(206) In some embodiments, the incorporation of a lattice structure into a striking face on a wood-type golf club head may enable the striking face to define a reduced thickness, for example, when compared to a striking face fabricated solely from a solid material, due to the stiffness provided by the lattice structure. That is, the incorporation of a lattice structure, or a ribbed structure (see FIG. **61**), on a striking face of a wood-type golf club head may provide added stiffness, which enables a thickness of the solid portion (i.e., a thickness defined by the portion of the striking face that does not include an added 3-D structure) to define a reduced thickness when compared to a striking face fabricated solely from solid material.

(207) Turning to FIG. **61**, in some embodiments, a face insert **810** may include of a wood-type golf club head may include a ribbed structure **812** arranged on a rear side of a striking face. In the illustrated embodiment, the ribbed structure **812** may include a solid portion **814** that defines a generally solid protrusion that protrudes from the rear side of the face insert **810** and a plurality of ribbed segments **816** that extend along the rear side of the striking face (e.g., generally in a sole-topline direction or a vertical direction from the perspective of FIG. **61**). The plurality of ribbed segments **816** may be spaced laterally (e.g., in a left-right direction from the perspective of FIG. **61**) along the rear side of the face insert **810**.

(208) In some embodiments, the 3-D structures incorporated onto the striking faces of wood-type golf club heads may be difficult to manufacture using conventional manufacturing processes.

Additive manufacturing processes may be leveraged to enable efficient and accurate manufacturing of these striking faces of wood-type golf club heads. For example, FIG. **62** illustrates a face insert **820** that is based on the face insert **810** and manufactured via an additive manufacturing process and may be used in a casting process. In some embodiments, the face insert **820** may be manufactured out of an investment casting material (e.g., wax) and may be manufactured via an additive manufacturing process. In some embodiments, conventional, non-additive manufacturing processes may not be able to create the 3-D structure arranged on the rear side of the striking faces described herein, for example, do to the presence of a lattice structure, an undercut, or a gap. Additive manufacturing may be leveraged to efficiently and accurately manufacture a face insert **820**. Once the face insert **820** is manufactured via an additive manufacturing process, the face insert **820** may be used to create a casting mold, or another mold (e.g., metal injection molding mold), of a striking face of a wood-type golf club head by shelling the mold with a slurry to form a shell. Once the shell has formed, the investment casting material (e.g., wax) may be burned out and metal may be poured into the cavity defined by the shell to form a casting of the face insert. As illustrated in FIG. **63**, the casting mold or other type of mold may be used to manufacture the striking face of a wood-type golf club head with an accurate representation of the desired 3-D structure arranged on the rear side of the striking face. The manufactured striking face illustrated in FIG. **63** may then be post-processed to conform to factory finish standard and may be attached to a club head body.

(209) In some embodiments, the additive manufacturing of a mold, or a structure that is used to make a mold in an investment casting process, may be used to manufacture iron-type golf club heads. For example, FIG. **64** illustrates a club head mold **830** for an iron-type golf club head that may be manufactured via an additive manufacturing process. In some embodiments, the club head **830** may be manufactured out of an investment casting material (e.g., wax). The manufacture of the club head **830** via an additive manufacturing process may enable the creation of unique undercuts and intricate geometries, for example, arranged on a rear surface or rear cavity of an iron-type golf club head, among other locations. For example, FIGS. **65** and **66** illustrate a 3-D structure that may be incorporated into the club head **830** of FIG. **64**.

(210) In general, the use of a wax pattern mold that is printed via an additive manufacturing process may increase efficiency, decrease costs, and enable the creation of more complex club head geometries, when compared to conventional manufacturing processes. For example, creating a wax pattern mold via an additive manufacturing process does not require tooling when creating a design of the mold. A 3-D model of the mold may be created in 3-D printing software, where conventional investment casting mold requires the creation of a wax tool. Once the part is designed in 3-D printing software, the wax pattern mold may be printed via an additive manufacturing process with casting gates, while conventional investment castings require wax to be injected into the wax tool.

(211) As described herein, in some embodiments, a golf club head may be required to be sintered after manufacture via an additive manufacturing process. In these embodiments, a support structure or fixture may be required to aid in maintaining orientation and shape of the green part during sintering. FIGS. **67** and **68** illustrate one embodiment of a sintering support **900** that may be used to support a golf club head during sintering. In the illustrated embodiment, the sintering support **900** may be used to support an iron-type golf club head during sintering. The sintering support **900** may include a face surface **902**, a hosel surface **904**, and a support wall **906**. The hosel surface **904** may extend from one side of the face surface **902** at an angle that is defined by a lie angle of the golf club head. The support wall **906** may extend generally perpendicularly from a side of the face surface **902** that is opposite to the hosel surface **904**.

(212) In the illustrated embodiment, the face surface **902** may be arranged generally parallel to a sintering plane S that the sintering support **900** rests on during sintering. In this way, for example, when a golf club head is arranged on the sintering support **900**, the face surface **902** orients a front face or striking face of a golf club head generally parallel to the build plane and provides support to

the front face or striking face. The arrangement and support of the front face or striking face provided by the sintering support **900** aids in reducing or preventing warping of the golf club head geometry during sintering. In addition, the angle between the face surface **902** and the hosel surface **904** being equal to a lie defined by the golf club head further aids in reducing or preventing warping of the golf club head geometry during sintering.

(213) Referring to FIGS. **69** and **70**, in some embodiments, the sintering support **900** may angle the golf club head supported thereon, such that a hosel of the golf club head is arranged generally perpendicular to the sintering plane S or generally parallel to a direction of gravity. In this way, for example, the sintering support **900** may further aid in preventing warping of the club head geometry during the sintering process. In the illustrated embodiment, a bottom edge **908** of the support wall **906** may be arranged at an angle relative to the face surface **902**. When the bottom edge **908** of the support wall **906** is placed on the sintering plane P, the angle between the bottom edge **908** of the support wall **906** and the face surface **902** may arrange a hosel **910** of a golf club head **912** in a direction that is generally perpendicular to the sintering plane P or generally parallel to a direction of gravity. In this orientation, the face surface **902** may be angled relative to the sintering plane S. The orientation of the hosel **910** in a direction that is generally perpendicular to a direction of gravity may prevent movement of the hosel **910** during sintering, which maintains the lie and loft defined by the golf club head **912** pre-sintering.

(214) In some embodiments, the sintering support **900** may be fabricated via an additive manufacturing process. For example, the face surface **902**, the hosel surface **904**, and the support wall **906** may be formed layer by layer by an additive manufacturing process.

(215) Any of the embodiments described herein may be modified to include any of the structures or methodologies disclosed in connection with different embodiments. Further, the present disclosure is not limited to club heads of the type specifically shown. Still further, aspects of the club heads of any of the embodiments disclosed herein may be modified to work with a variety of golf clubs.

(216) As noted previously, it will be appreciated by those skilled in the art that while the disclosure has been described above in connection with particular embodiments and examples, the disclosure is not necessarily so limited, and that numerous other embodiments, examples, uses, modifications and departures from the embodiments, examples and uses are intended to be encompassed by the claims attached hereto. Various features and advantages of the invention are set forth in the following claims.

INDUSTRIAL APPLICABILITY

(217) Numerous modifications to the present disclosure will be apparent to those skilled in the art in view of the foregoing description. Accordingly, this description is to be construed as illustrative only and is presented for the purpose of enabling those skilled in the art to make and use the invention. The exclusive rights to all modifications which come within the scope of the appended claims are reserved.

Claims

1. A golf club head, comprising: a body including a topline, a sole, and a front face, wherein the front face defines a rear surface that extends along a plane and the body defines a center of gravity plane, wherein the center of gravity plane extends parallel to a ground plane at a center of gravity of the body when the golf club head is at address; and a lattice structure formed on a portion of the body, layer by layer, via an additive manufacturing process, wherein the portion of the body is bounded by the plane, the center of gravity plane, the topline, and an intersection between the plane and the center of gravity plane, wherein the entire lattice structure is disposed above the center of gravity plane and behind the rear surface, wherein the lattice structure extends along the topline toward a toe end of the body, and wherein a thickness is measured normal to the front face between the front face and a back of the lattice structure, the thickness being smaller in a region proximate

the topline and larger in a region proximate the center of gravity plane.

2. The golf club head of claim 1, wherein the body is fabricated via the additive manufacturing process, and wherein the additive manufacturing process is a binder jetting process.

3. The golf club head of claim 1, wherein the lattice structure forms a part of the body.

4. The golf club head of claim 1, wherein, in an assembled configuration, the lattice structure is visible from an exterior of the golf club head.

5. The golf club head of claim 1, wherein the lattice structure includes a plurality of unit cells that vary in at least one of a unit cell size or a unit cell shape.

6. The golf club head of claim 5, wherein an upper region of the body is entirely above the center of gravity plane, a lower region of the body is entirely below the center of gravity plane, and the lower region has a density that is greater than a density of the upper region.

7. A golf club head, comprising: a body including a topline, a sole, and a front face, wherein the body defines a center of gravity plane, wherein the center of gravity plane is defined as a plane that extends parallel to a ground plane at a center of gravity of the body when the golf club head is at address; and a lattice structure formed within a portion of the body, layer by layer, via an additive manufacturing process, wherein the center of gravity plane divides the body into a first region that includes the topline and a second region that includes the sole, wherein the lattice structure is arranged in the first region above the center of gravity plane and adjacent a toe end, wherein the second region of the body is below the center of gravity plane and has a density that is greater than a density of the first region, wherein the lattice structure includes a plurality of interior voids that are in fluid communication with at least one void that is formed through an exterior surface on a toe end of the golf club head, and wherein, in an assembled configuration, the lattice structure is visible from an exterior of the golf club head.

8. The golf club head of claim 7, wherein the front face defines a rear surface that extends along a rear plane, and wherein the portion of the body is bounded by the rear plane, the center of gravity plane, and an intersection between the rear plane and the center of gravity plane.

9. The golf club head of claim 8, wherein the entire body above the center of gravity plane and rearward of the rear plane includes the lattice structure.

10. The golf club head of claim 7, wherein the lattice structure and the body are formed from the same metal and are integral with one another.

11. The golf club head of claim 7, wherein at least a portion of the lattice structure is covered by a solid material of the body.

12. A golf club head, comprising: a body including a topline, a sole, and a front face, wherein the body defines a center of gravity plane, wherein the center of gravity plane is defined as a plane that extends parallel to a ground plane at a center of gravity of the body when the golf club head is at address; and a lattice structure formed within a portion of the body, layer by layer, via an additive manufacturing process, wherein the lattice structure is arranged above the center of gravity plane and extends along an exterior of the topline from a heel end to a toe end of the body, wherein, in an assembled configuration, a portion of the lattice structure is visible from an exterior of the golf club head, and wherein a thickness is measured normal to the front face between the front face and a back of the lattice structure, the thickness being smaller in a region proximate the topline and larger in a region proximate the center of gravity plane.

13. The golf club head of claim 12, wherein the front face defines a rear surface that extends along a rear plane, and wherein the portion of the body is bounded by the rear plane, the center of gravity plane, and an intersection between the rear plane and the center of gravity plane.

14. The golf club head of claim 13, wherein an entire region of the body that is disposed above the center of gravity plane and rearward of the rear plane includes the lattice structure.

15. The golf club head of claim 12, wherein the lattice structure and the body are formed from the same metal and are integral with one another.

16. The golf club head of claim 12, wherein the lattice structure is visible to a user when the golf

club head is at address.

17. The golf club head of claim 12, wherein the lattice structure includes a plurality of unit cells that vary in at least one of a unit cell size or a unit cell shape.
